Learning a Codebase

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Contents

1 Repository

1.1 Glossary - Summary

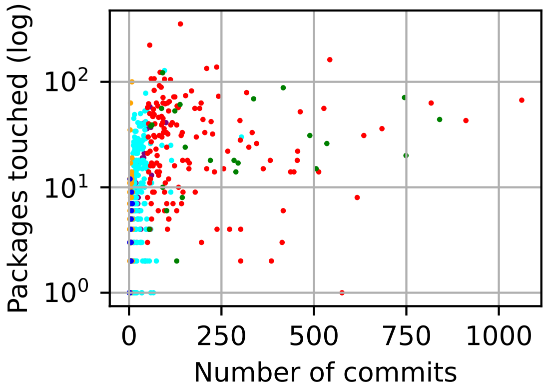
- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Repository Summary Table

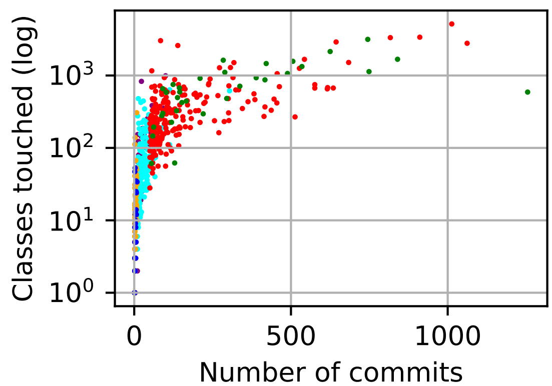
Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least zero sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	**Repo Name**	**Transient Founder**	**Moderate Founder**	**Sustained Founder**	**Transient Joiner**	**Moderate Joiner**	**Sustained Joiner**	**Commit**	**Start**	**End**
3	activiti	6	7	6	212	20	5	8662	2010-Jun-18	On going
5	airbyte-platform	16	2	1	457	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	129	32	15	4247	2011-Sep-22	On going
10	automq	4	0	0	301	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	18	5	6	531	85	21	14876	2013-Mar-21	2022-May-05
15	camel	1	4	1	1146	63	13	36528	2007-Mar-19	On going
18	checkstyle	0	0	1	362	42	9	17338	2001-Jun-22	On going
20	cxf	6	6	3	201	17	3	7893	2008-Apr-29	On going
21	intellij-community	0	3	7	344	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	402	8	3	9685	2009-Sep-15	On going
28	jenkins	5	1	1	837	39	4	30252	2006-Nov-05	On going
33	openmrs-core	0	1	0	388	26	3	15757	2006-May-03	On going
36	presto	0	1	3	692	71	18	30605	2012-Aug-09	On going
37	quarkus	2	3	5	824	25	8	81612	2018-Jun-14	On going
39	selenium	5	3	0	712	17	8	23354	2004-Nov-03	On going
	Total	63	40	34	7538	580	203	390714		

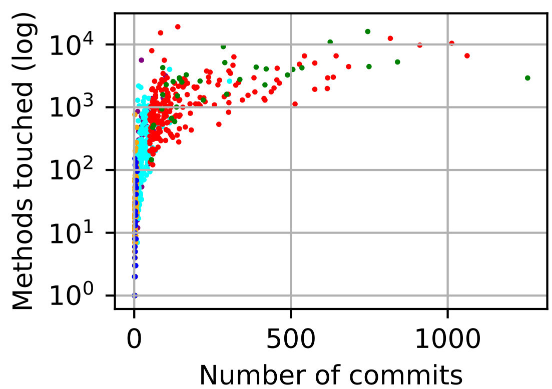
1.3 Sample scatter touched against commit -



(a) Packages



(b) Classes



(c) Methods

Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 sustained later joiners from six (6) categories: Transient Founder (Blue, 63), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

1.4 Total commits by developer - Developer Commits

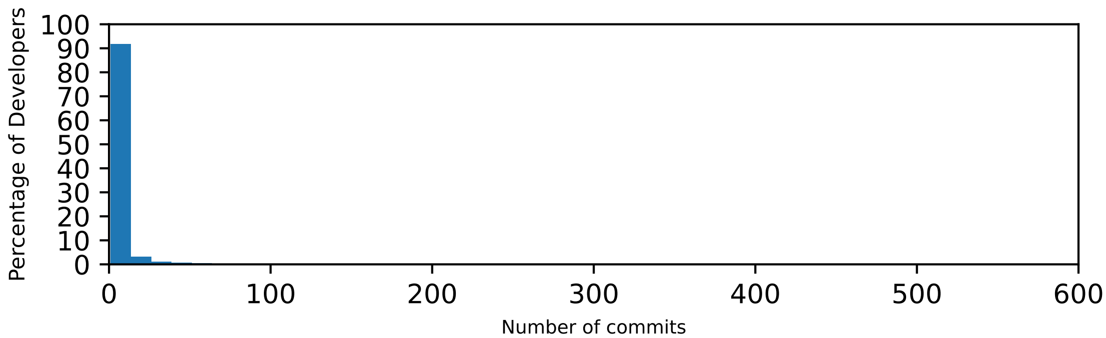
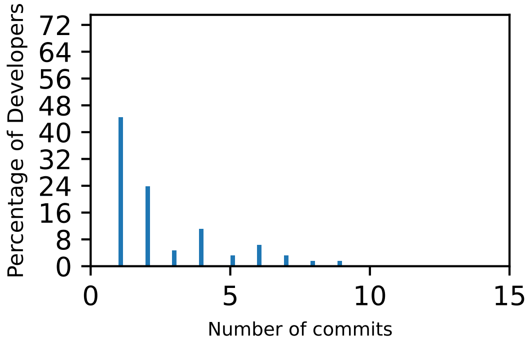
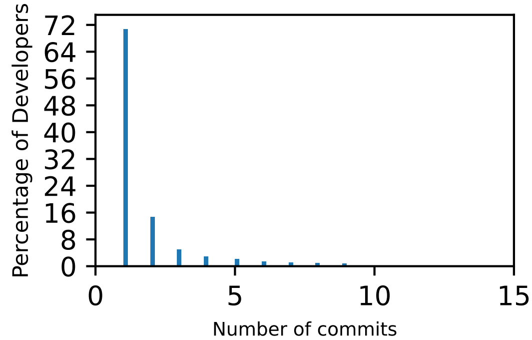


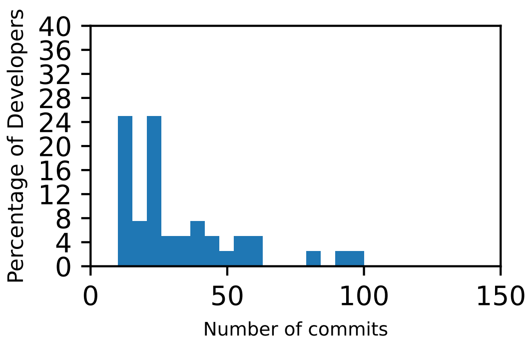
Figure 2: Histogram of percentage of all developers (n=8440) (y-axis) against the total number of commits (x-axis) in 15 repositories sampled from GitHub.



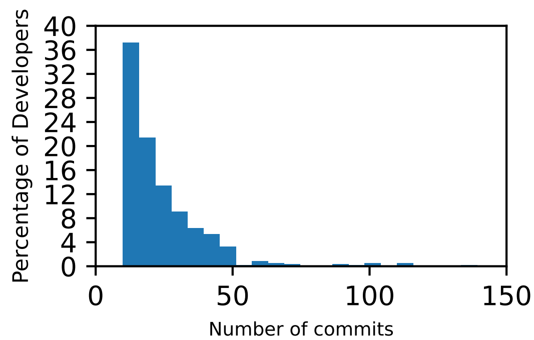
(a) Transient founder (n=63).



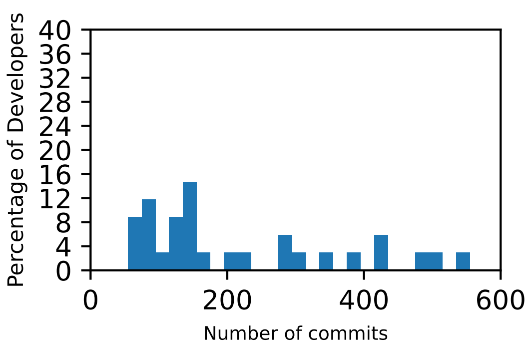
(b) Transient later joiner (n=7520).



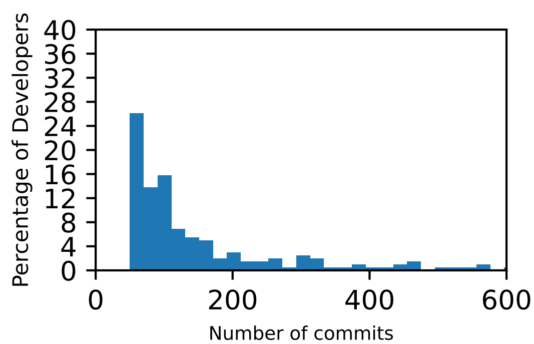
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



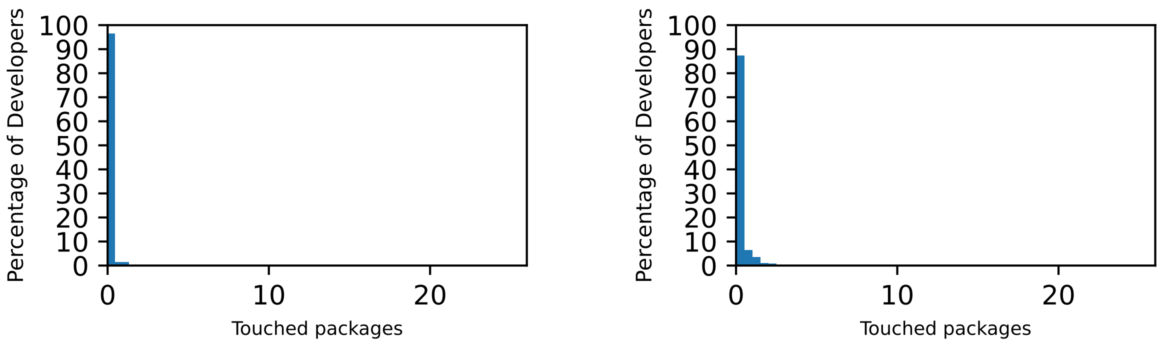
(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 3: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

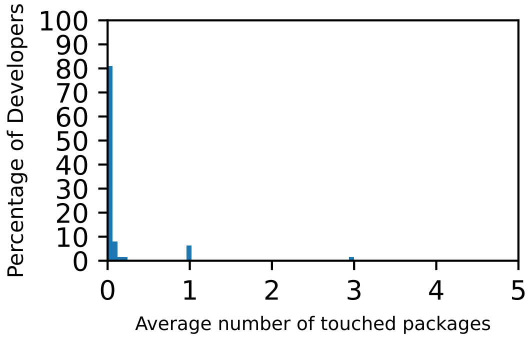
1.5 Histogram components touched by developer - For components with total



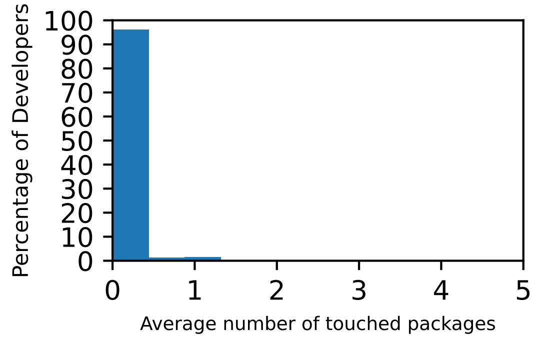
(a) By Day.

(b) By Commit.

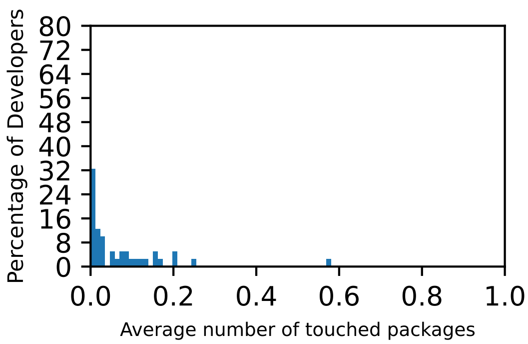
Figure 4: Histogram of percentage of all developers (y-axis) against the average packages touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



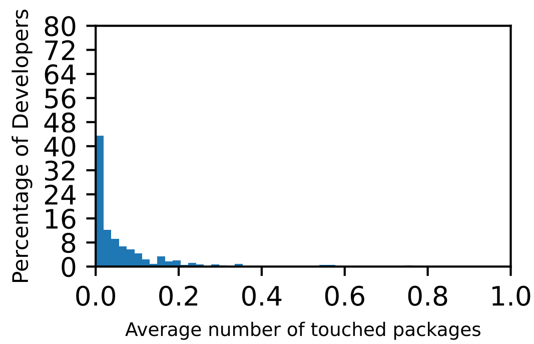
(a) Transient founder (n=63).



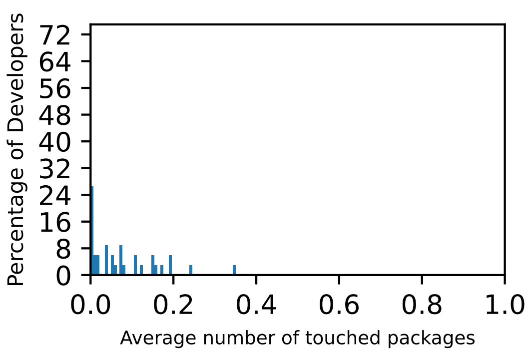
(b) Transient later joiner (n=7520).



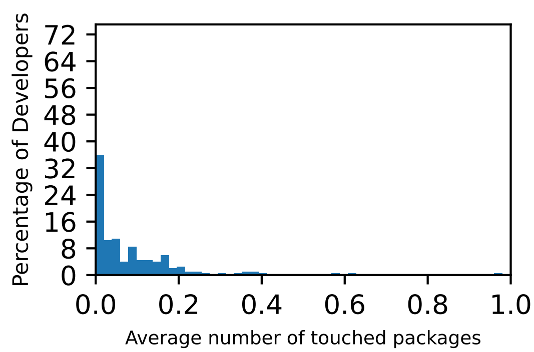
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

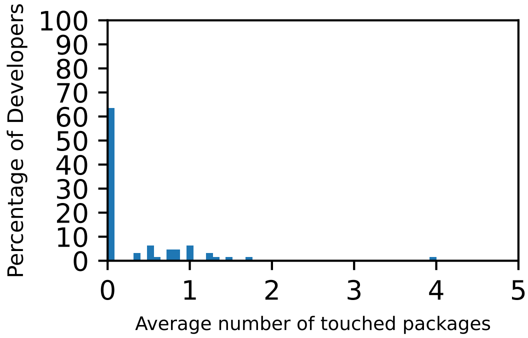


(e) Sustained founder (n=34).

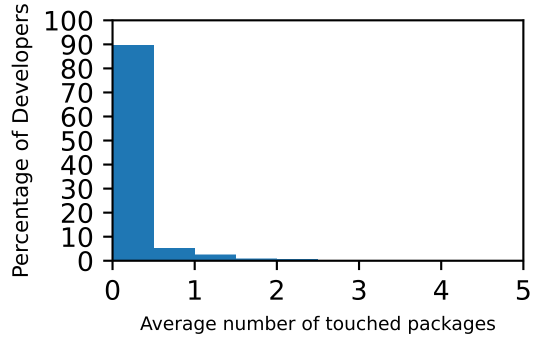


(f) Sustained later joiner (n=203).

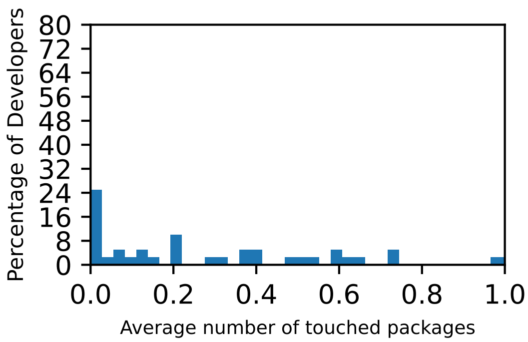
Figure 5: Histogram of percentage of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



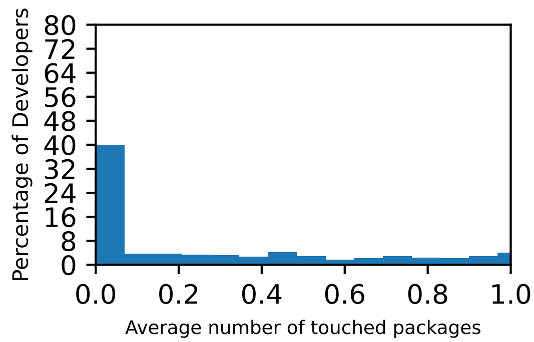
(a) Transient founder (n=63).



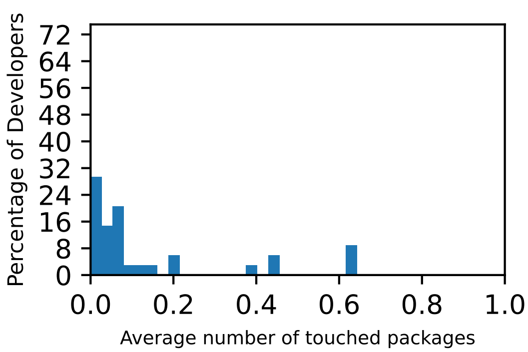
(b) Transient later joiner (n=7520).



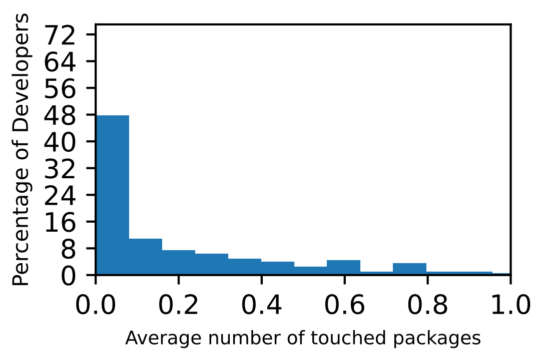
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

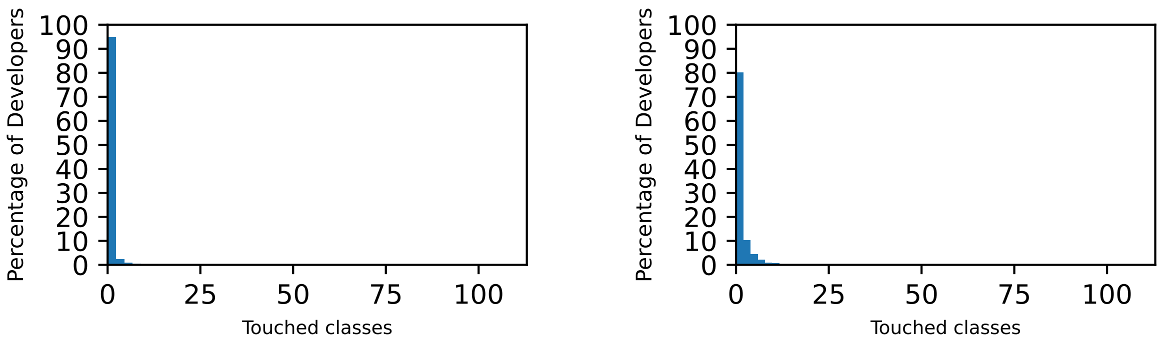


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

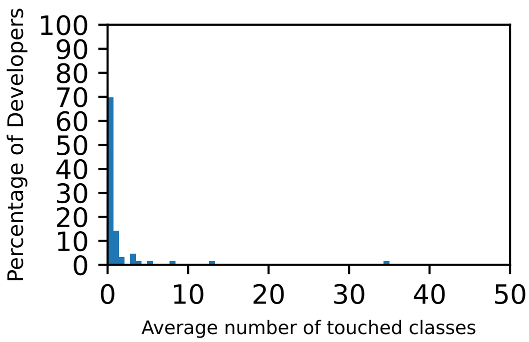
Figure 6: Histogram of percentage of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



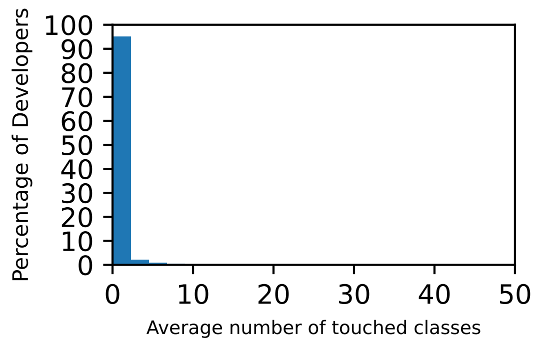
(a) By Day.

(b) By Commit.

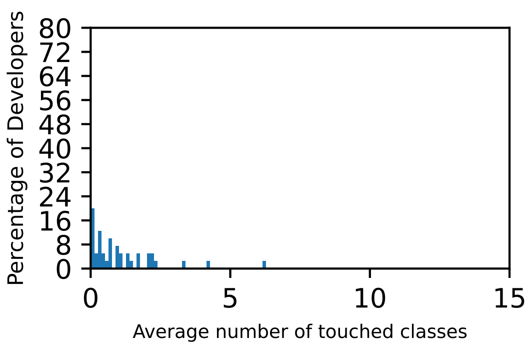
Figure 7: Histogram of percentage of all developers (y-axis) against the average classes touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



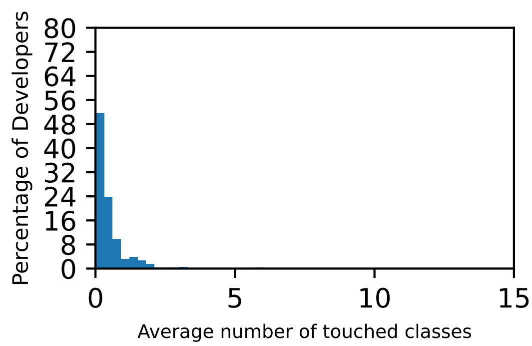
(a) Transient founder (n=63).



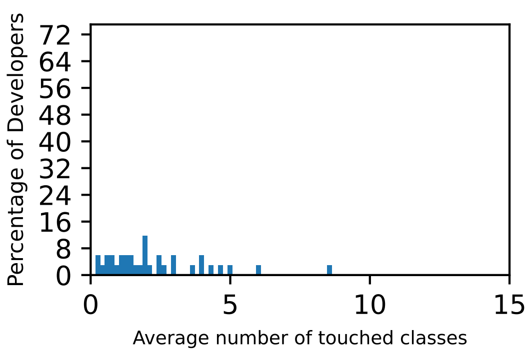
(b) Transient later joiner (n=7520).



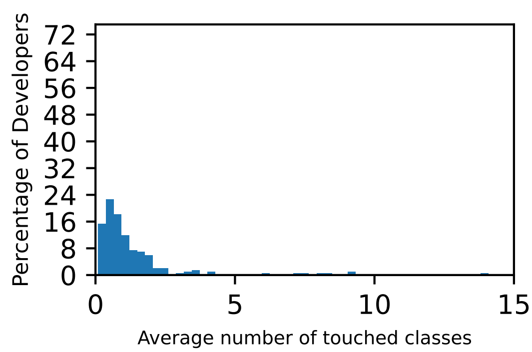
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

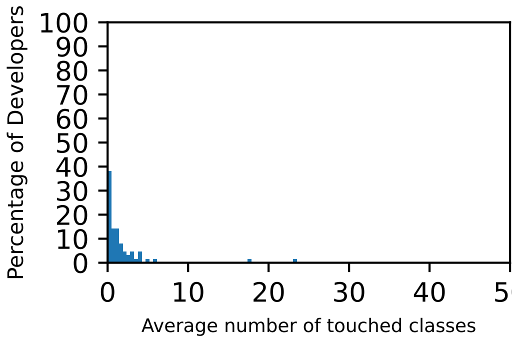


(e) Sustained founder (n=34).

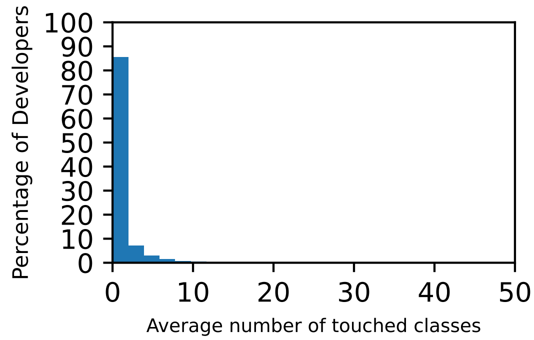


(f) Sustained later joiner (n=203).

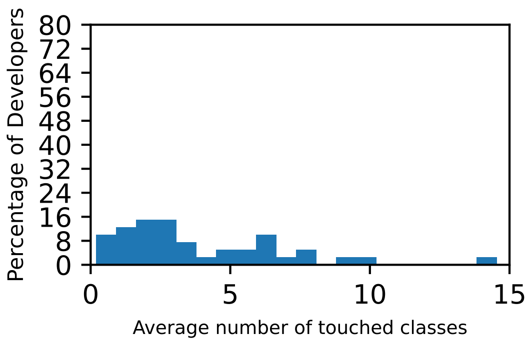
Figure 8: Histogram of percentage of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



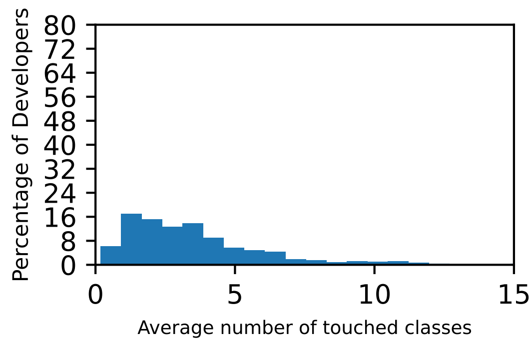
(a) Transient founder (n=63).



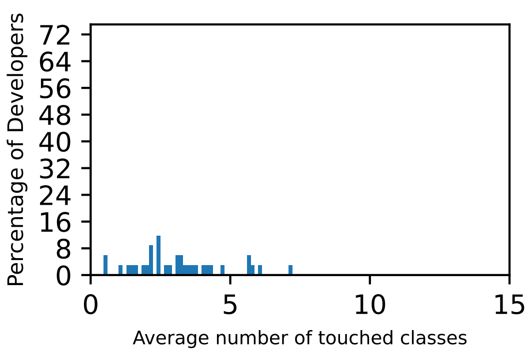
(b) Transient later joiner (n=7520).



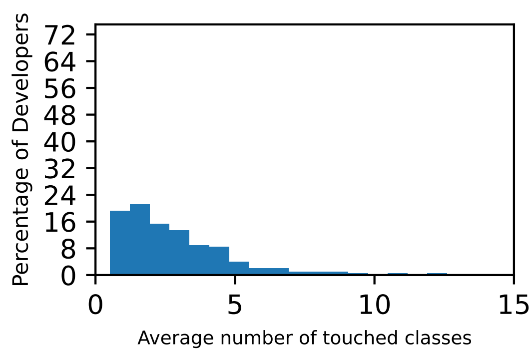
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

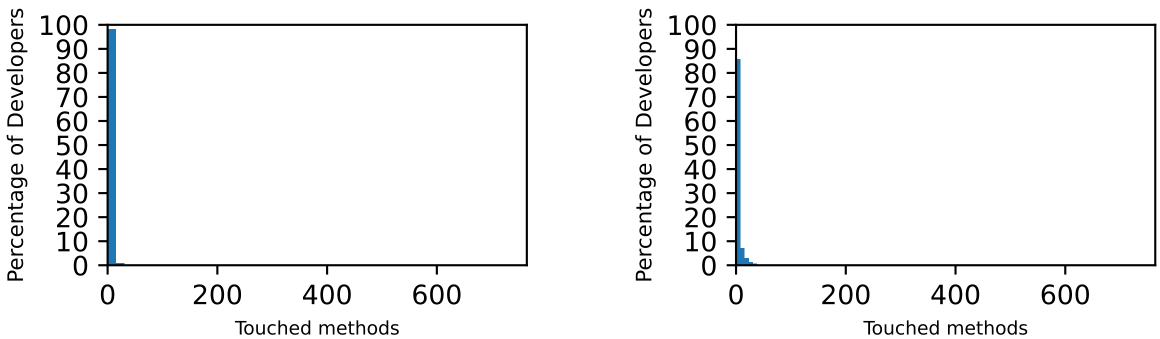


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

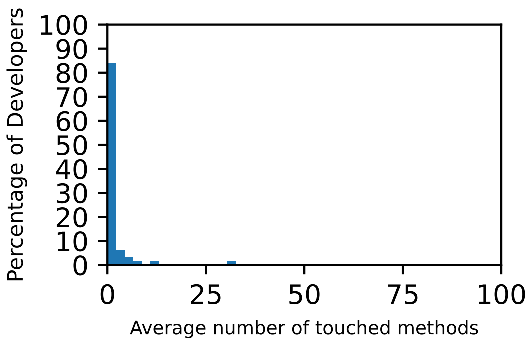
Figure 9: Histogram of percentage of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



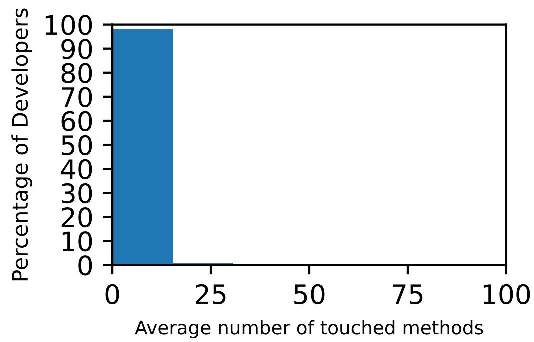
(a) By Day.

(b) By Commit.

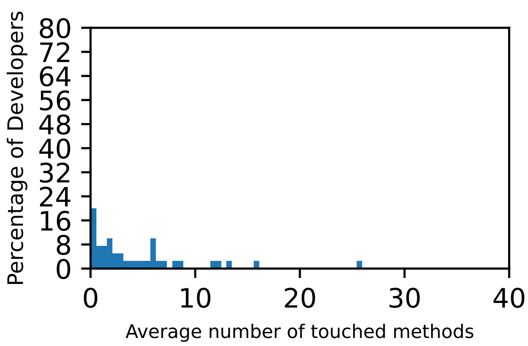
Figure 10: Histogram of percentage of all developers (y-axis) against the average methods touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



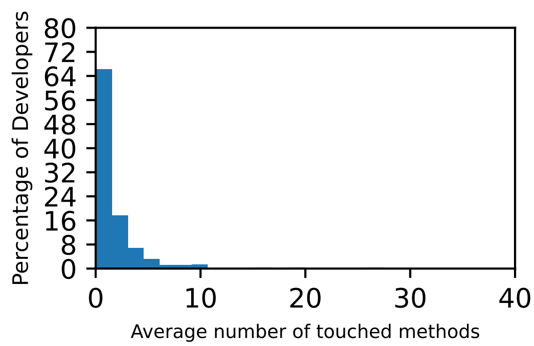
(a) Transient founder (n=63).



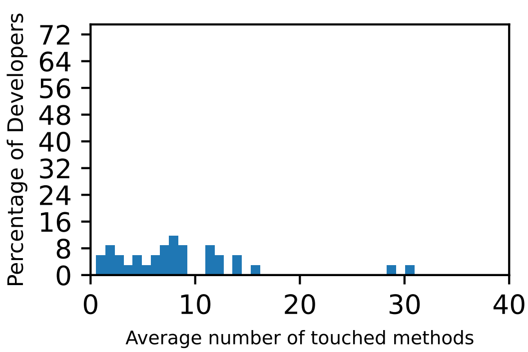
(b) Transient later joiner (n=7520).



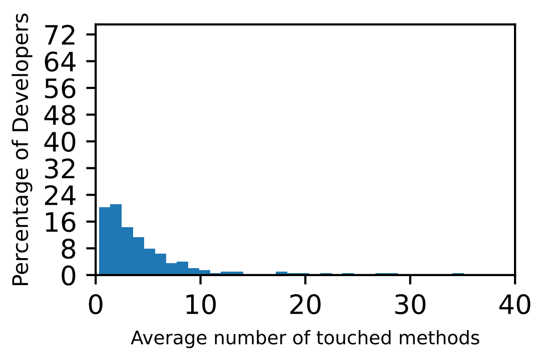
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

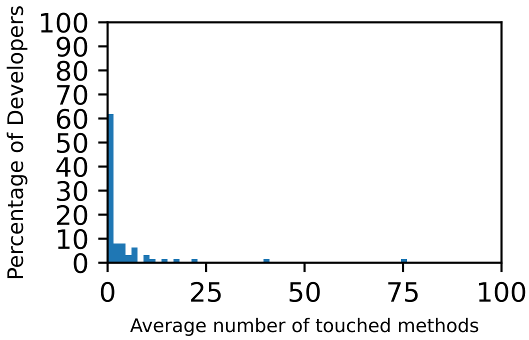


(e) Sustained founder (n=34).

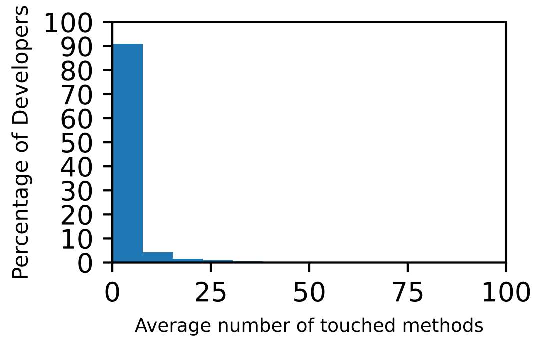


(f) Sustained later joiner (n=203).

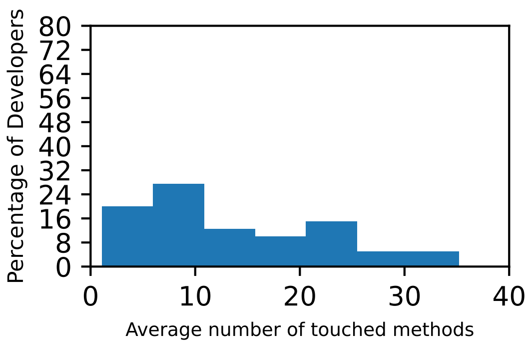
Figure 11: Histogram of percentage of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



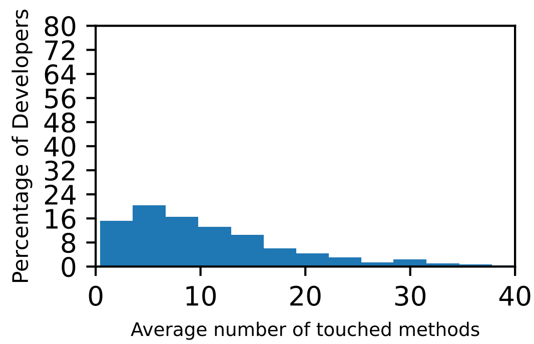
(a) Transient founder (n=63).



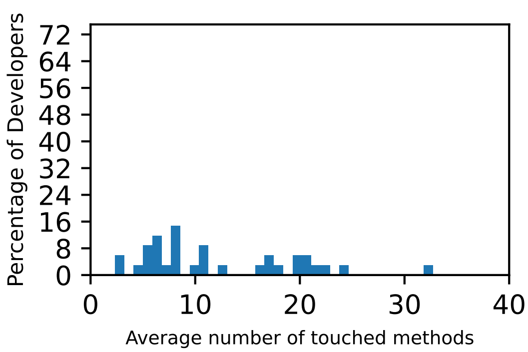
(b) Transient later joiner (n=7520).



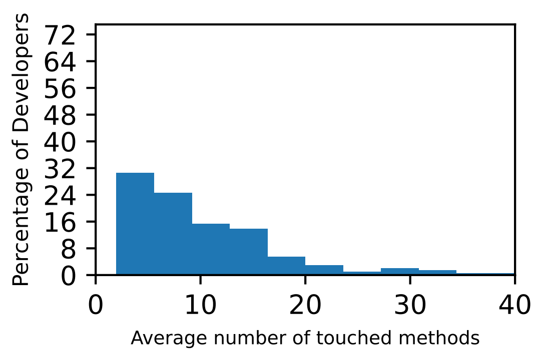
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



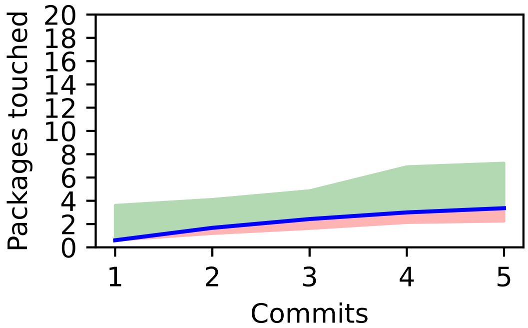
(e) Sustained founder (n=34).



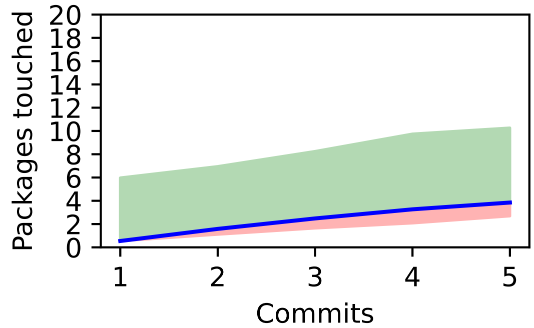
(f) Sustained later joiner (n=203).

Figure 12: Histogram of percentage of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

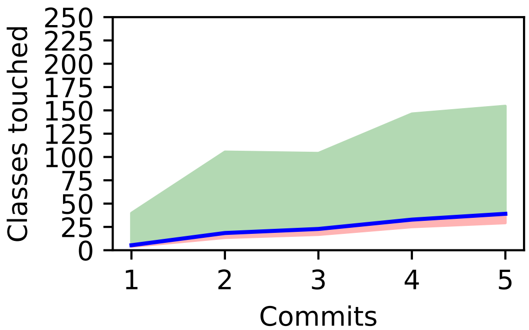
1.5.1 Time series components touched - For packages touched for first five commits



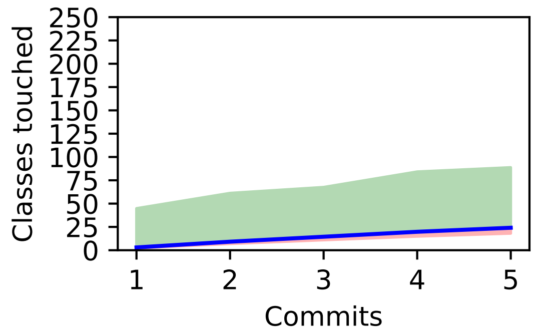
(a) Packages for All founders (n=74 & μ=3.36)



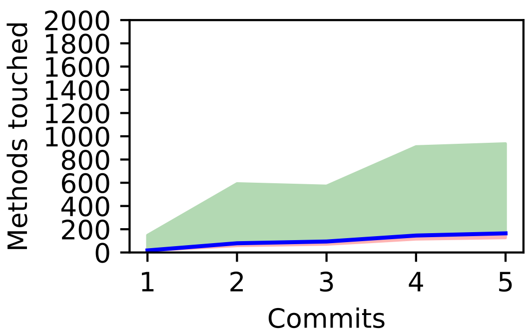
(b) Packages for All joiners (n=783 & μ=3.85)



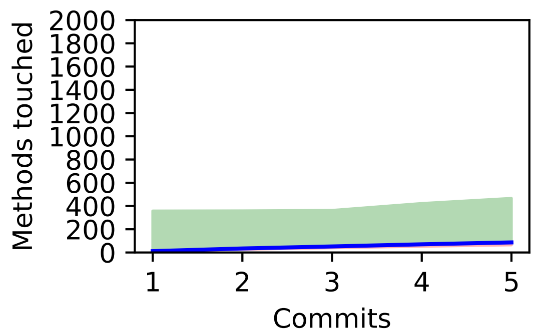
(c) Classes for All founders (n=74 & μ=39.05)



(d) Classes for All joiners (n=783 & μ=24.11)

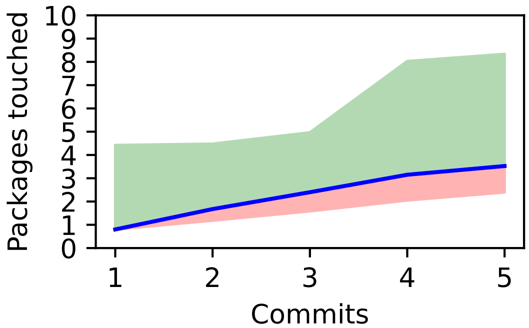


(e) Methods for All founders (n=74 & μ=164.59)

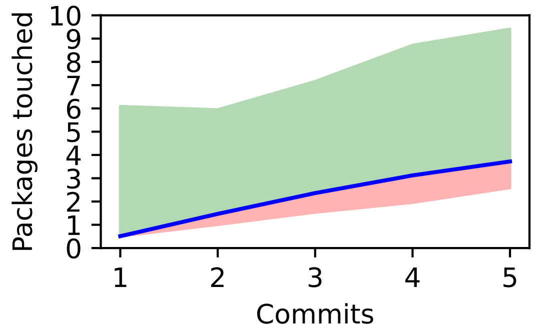


(f) Methods for All joiners (n=783 & μ=86.68)

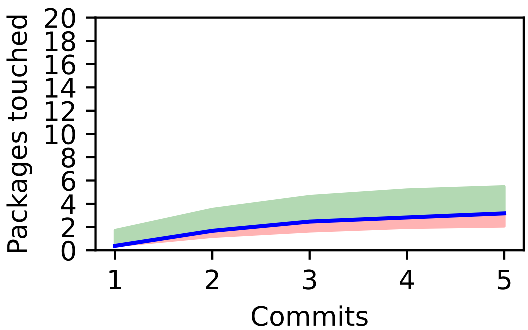
Figure 13: A time series for the five (5) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



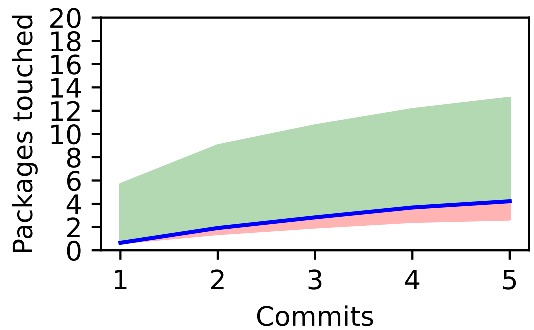
(a) Moderate founder (n=40 & μ=3.52)



(b) Moderate later joiner (n=580 & μ=3.72)

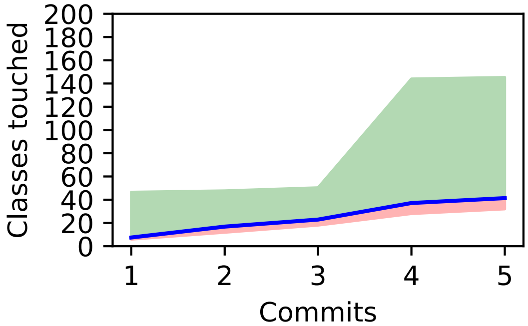


(c) Sustained founder (n=34 & μ=3.18)

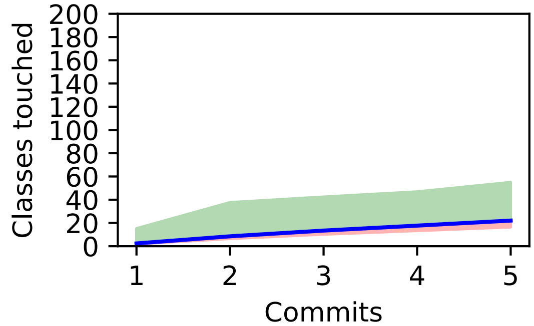


(d) Sustained later joiner (n=203 & μ=4.22)

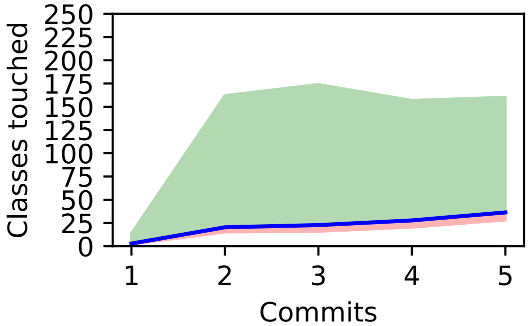
Figure 14: A time series for the five (5) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



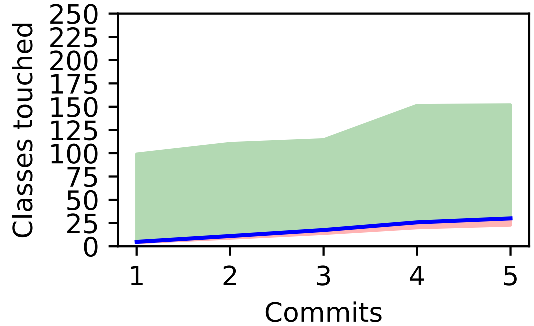
(a) Moderate founder (n=40 & μ=41.38)



(b) Moderate later joiner (n=580 & μ=22.04)



(c) Sustained founder (n=34 & μ=36.32)



(d) Sustained later joiner (n=203 & μ=30.01)

Figure 15: A time series for the five (5) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

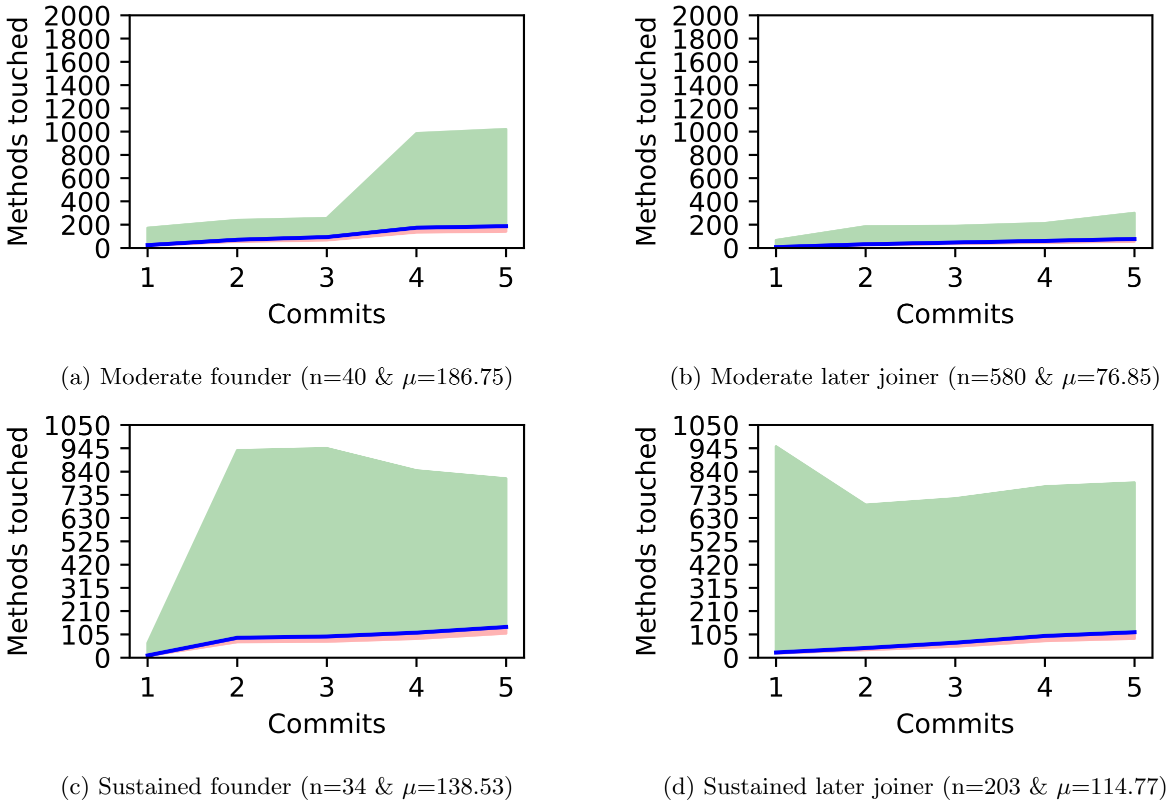
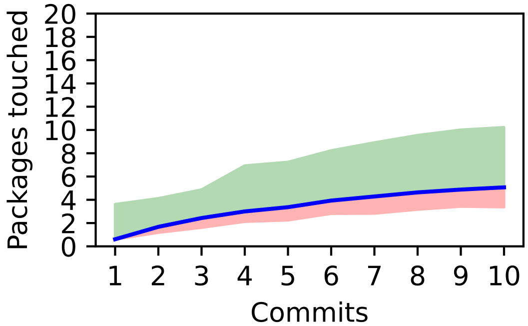
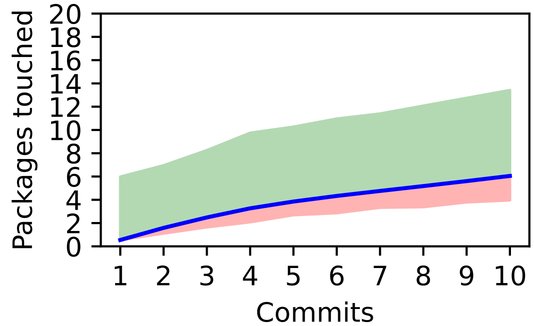


Figure 16: A time series for the five (5) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

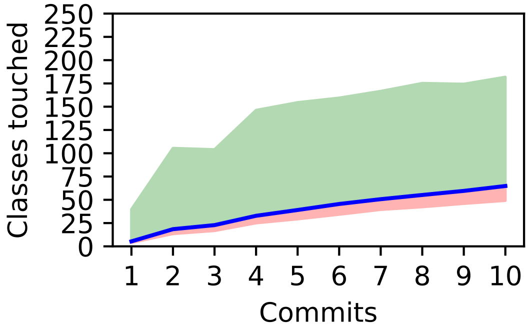
1.5.2 Time series components touched - For packages touched for first ten commits



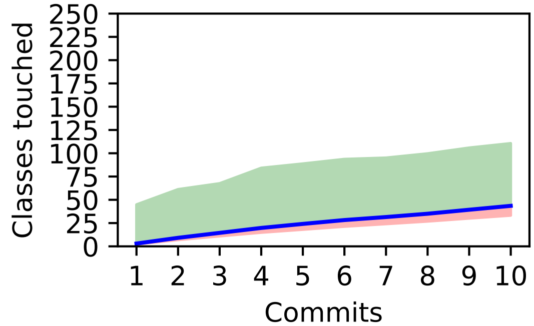
(a) Packages for All founders (n=74 & μ=5.07)



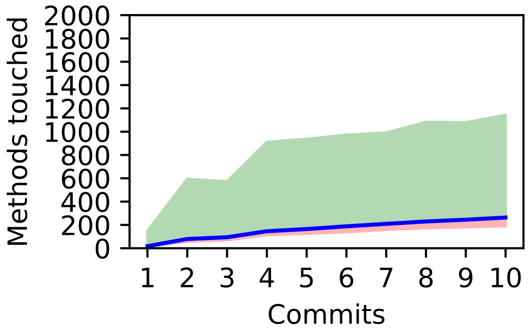
(b) Packages for All joiners (n=783 & μ=6.05)



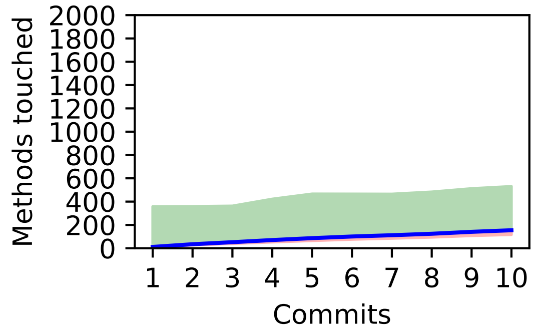
(c) Classes for All founders (n=74 & μ=64.81)



(d) Classes for All joiners (n=783 & μ=43.51)

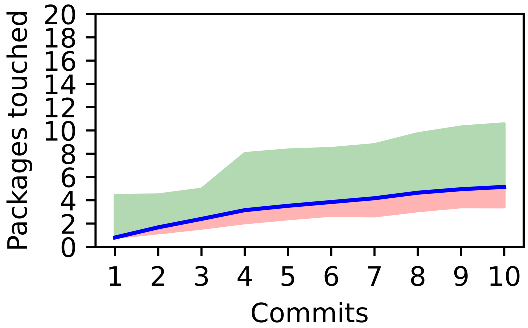


(e) Methods for All founders (n=74 & μ=263.26)

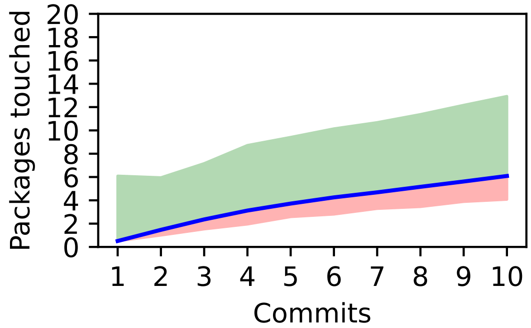


(f) Methods for All joiners (n=783 & μ=154.41)

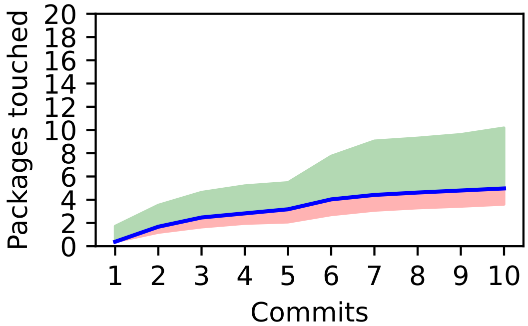
Figure 17: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



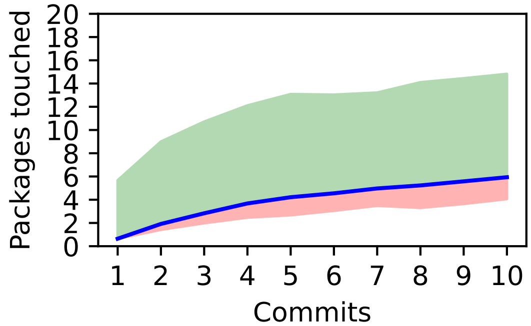
(a) Moderate founder (n=40 & μ=5.15)



(b) Moderate later joiner (n=580 & μ=6.09)

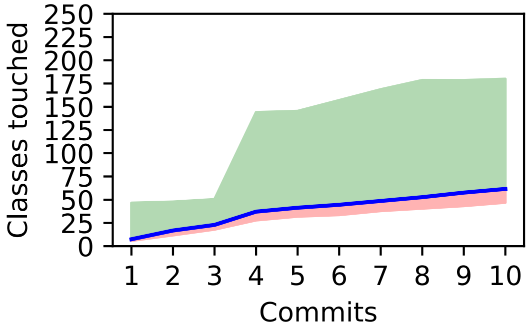


(c) Sustained founder (n=34 & μ=4.97)

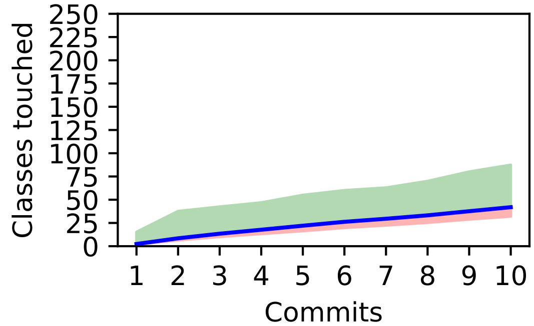


(d) Sustained later joiner (n=203 & μ=5.94)

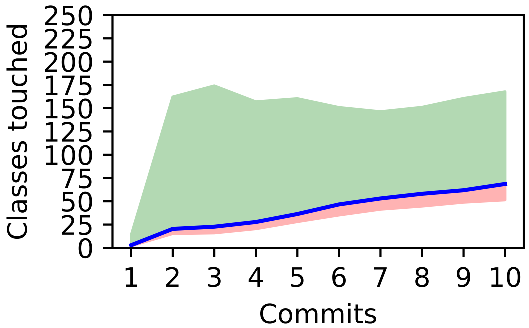
Figure 18: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



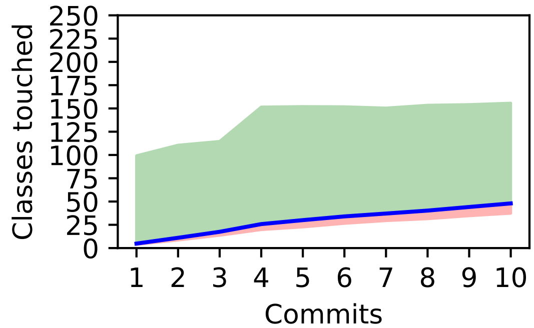
(a) Moderate founder (n=40 & μ=61.6)



(b) Moderate later joiner (n=580 & μ=41.95)

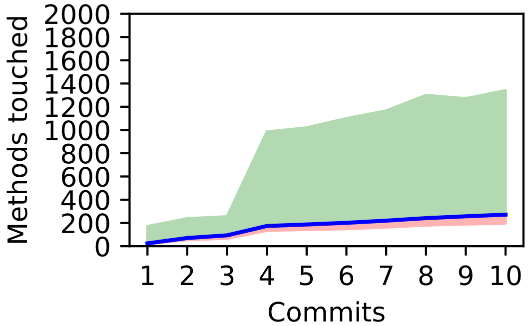


(c) Sustained founder (n=34 & μ=68.59)

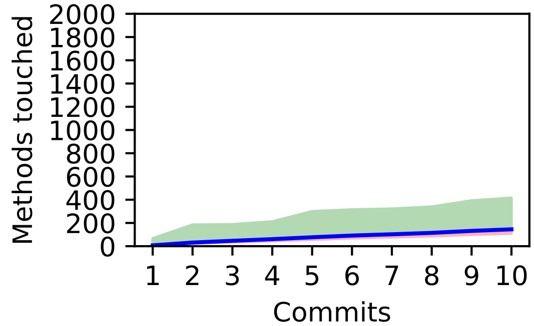


(d) Sustained later joiner (n=203 & μ=47.98)

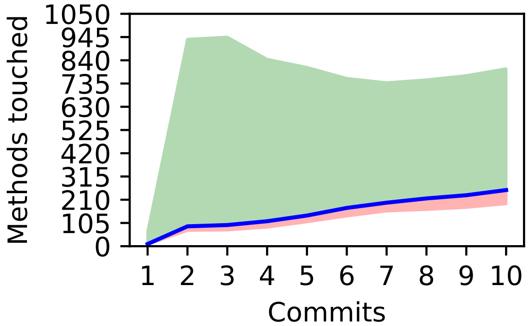
Figure 19: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



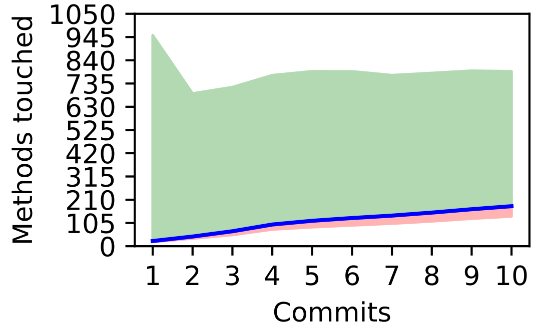
(a) Moderate founder (n=40 & μ=271.42)



(b) Moderate later joiner (n=580 & μ=145.21)



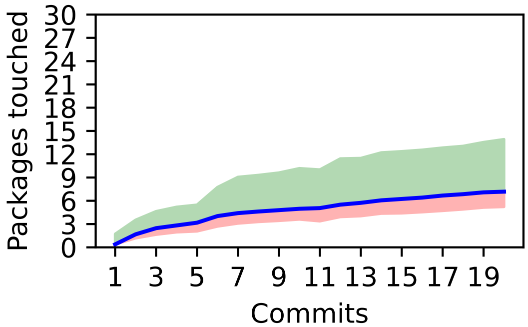
(c) Sustained founder (n=34 & μ=253.65)



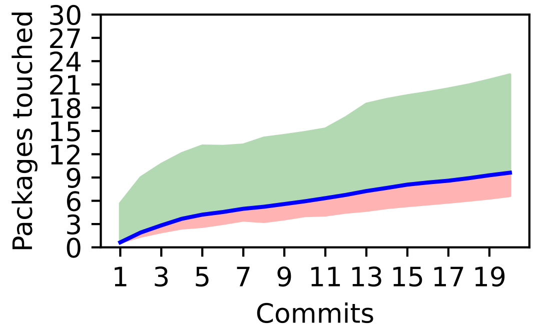
(d) Sustained later joiner (n=203 & μ=180.7)

Figure 20: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

1.5.3 Time series components touched - For packages touched for first twenty commits

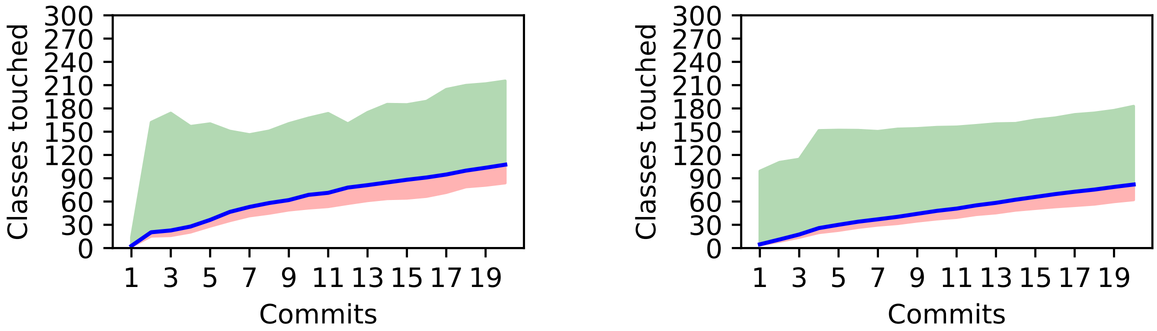


(a) Sustained founder (n=34 & μ=7.18)



(b) Sustained later joiner (n=203 & μ=9.62)

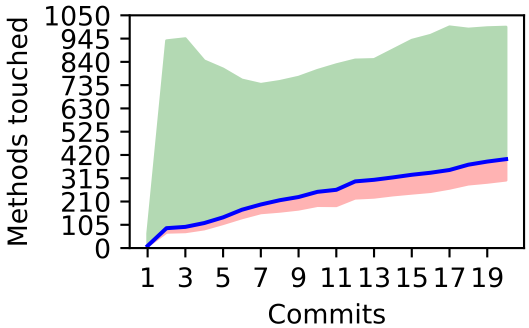
Figure 21: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



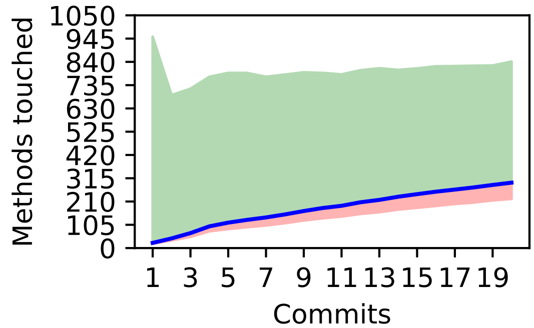
(a) Sustained founder (n=34 & μ=107.5)

(b) Sustained later joiner (n=203 & μ=81.91)

Figure 22: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



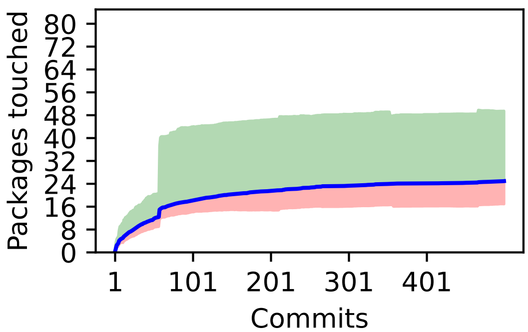
(a) Sustained founder (n=34 & μ=401.24)



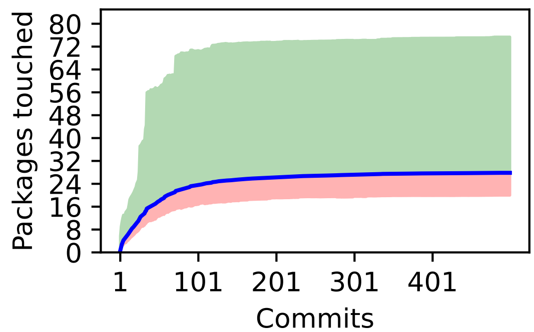
(b) Sustained later joiner (n=203 & μ=295.1)

Figure 23: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

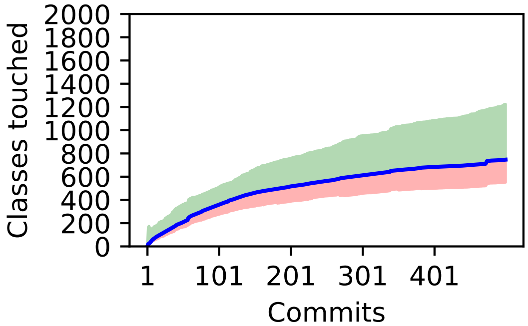
1.5.4 Time series components touched - For packages touched for commit



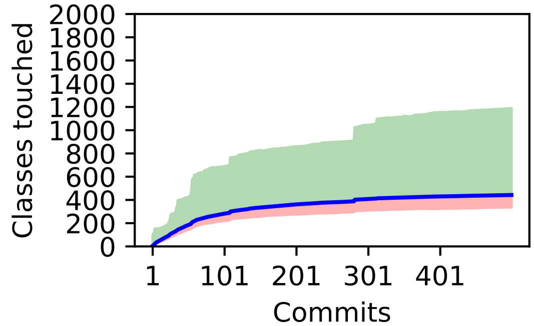
(a) Packages for All founders (n=34 & μ=24.94)



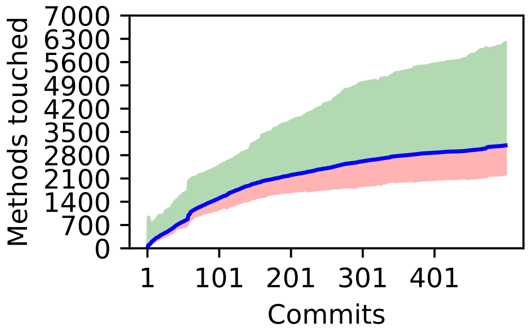
(b) Packages for All joiners (n=203 & μ=27.84)



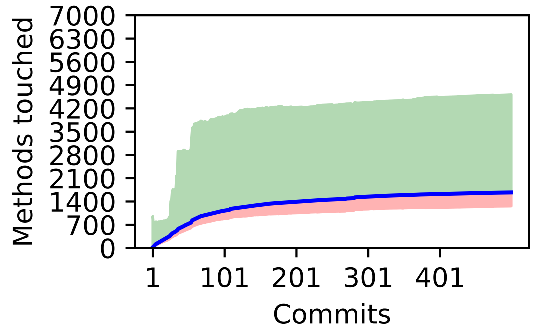
(c) Classes for All founders (n=34 & μ=746.59)



(d) Classes for All joiners (n=203 & μ=442.33)

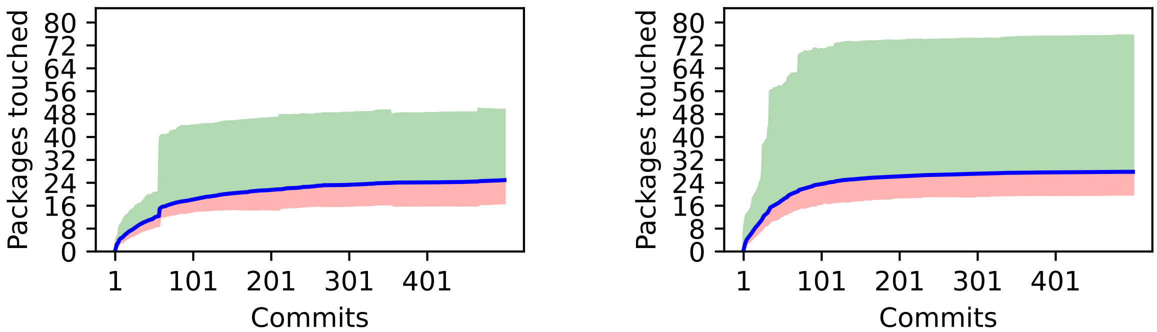


(e) Methods for All founders (n=34 & μ=3095.18)



(f) Methods for All joiners (n=203 & μ=1672.84)

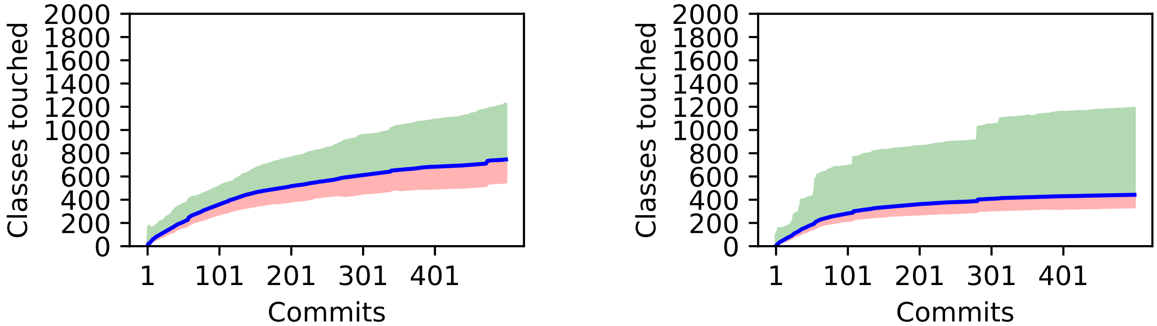
Figure 24: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34 & μ=24.94)

(b) Sustained later joiner (n=203 & μ=27.84)

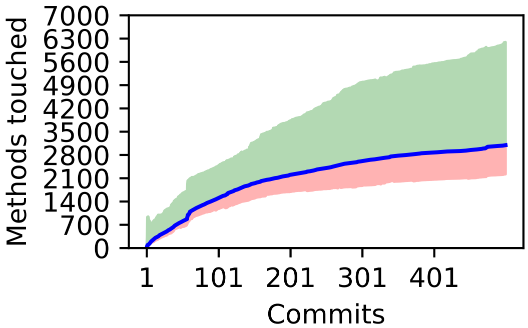
Figure 25: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



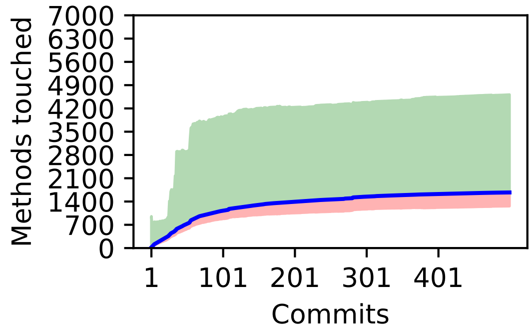
(a) Sustained founder (n=34 & μ=746.59)

(b) Sustained later joiner (n=203 & μ=442.33)

Figure 26: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



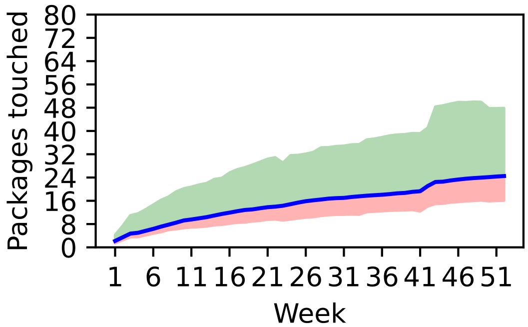
(a) Sustained founder (n=34 & μ=3095.18)



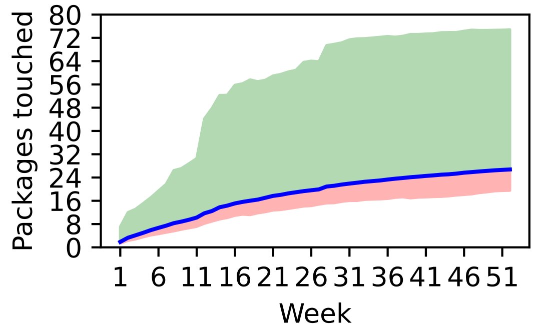
(b) Sustained later joiner (n=203 & μ=1672.84)

Figure 27: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

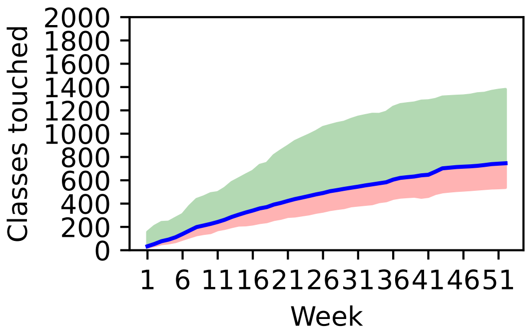
1.5.5 Time series components touched - For packages touched for each week commit



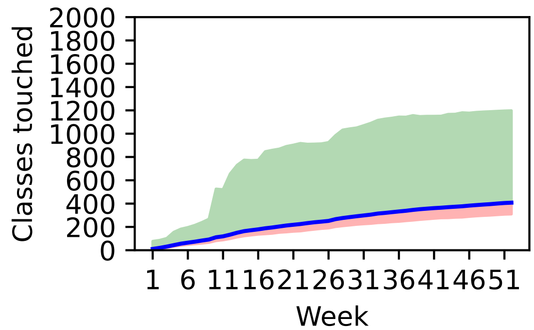
(a) Packages for All founders (n=34 & μ=24.5)



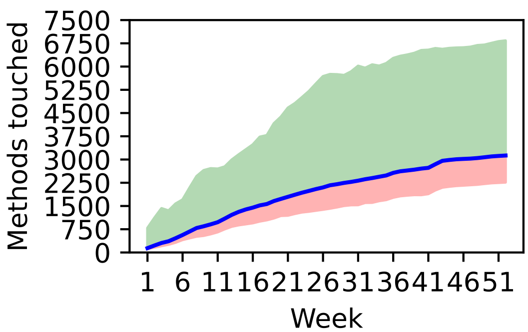
(b) Packages for All joiners (n=203 & μ=26.77)



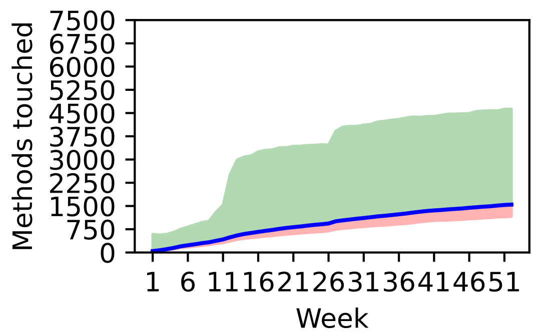
(c) Classes for All founders (n=34 & μ=746.15)



(d) Classes for All joiners (n=203 & μ=408.05)

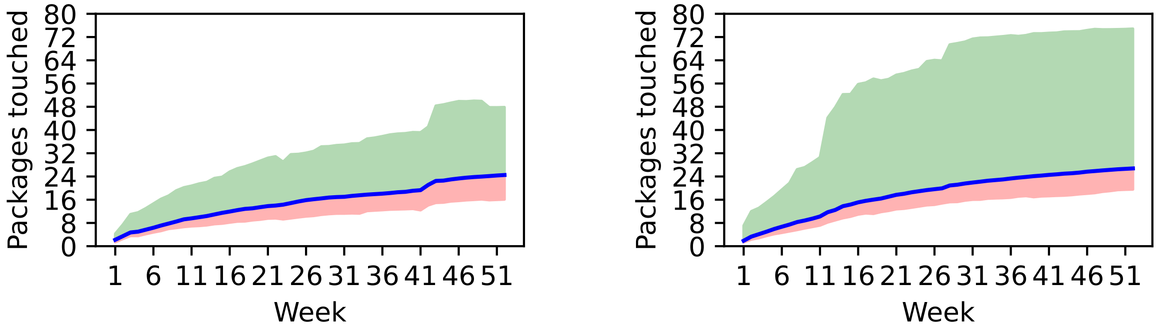


(e) Methods for All founders (n=34 & μ=3129.0)



(f) Methods for All joiners (n=203 & μ=1545.61)

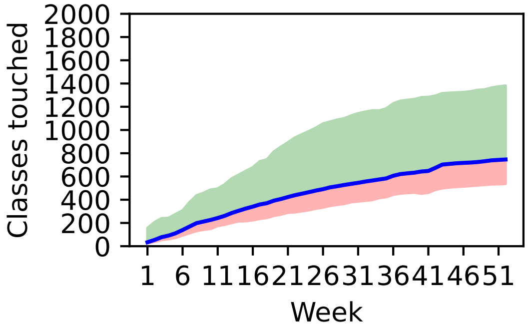
Figure 28: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



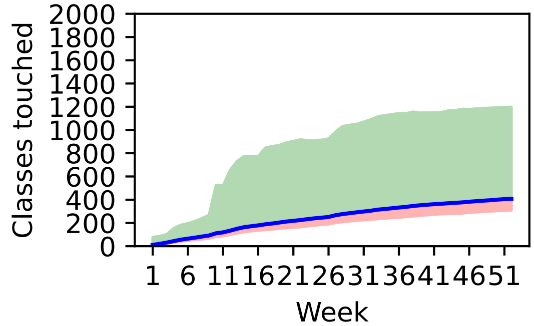
(a) Sustained founder (n=34 & μ=24.5)

(b) Sustained later joiner (n=203 & μ=26.77)

Figure 29: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

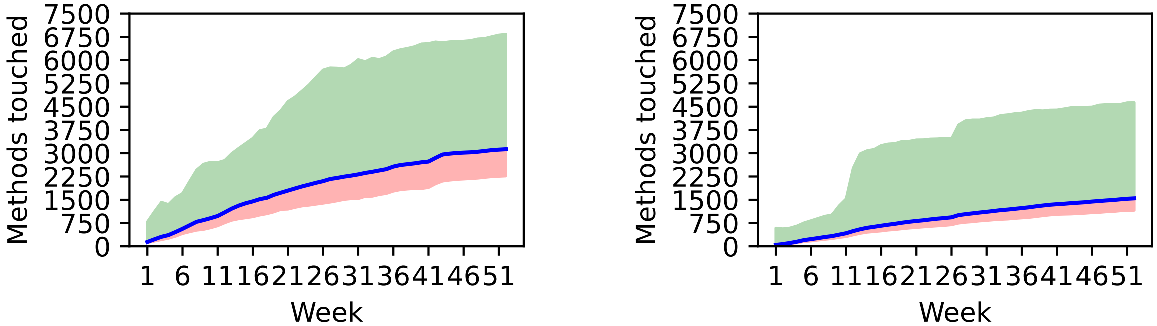


(a) Sustained founder (n=34 & μ=746.15)



(b) Sustained later joiner (n=203 & μ=408.05)

Figure 30: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34 & μ=3129.0)

(b) Sustained later joiner (n=203 & μ=1545.61)

Figure 31: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

1.6 Welch t test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances. Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers. Columns below:

- **Sample A** - F - Founder, LJ - Late Joiner, T - Transient, M - Moderate, S Sustained. In brackets number of developers.
- **Sample B** - F - Founder, LJ - Late Joiner, T - Transient, M - Moderate, S Sustained. In brackets number of developers.
- **Component** - Packages, classes or method touched.
- **Number of first commits** - A sample of the first x number of commits for developers.
- **Sample A** μ - The mean average of the first sample A.
- **Sample B** μ - The mean average of the second sample B.
- **Welch Statisic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** P - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU Statsitcal (Number of Pairs)** - The Mann-Whitney U Statistic represents the number of pairwise comparisons where one group outranks the other, with the total possible pairs provided in brackets.
- **MWU** P - The Mann-Whitney p-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.

Table 2: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of component (package, class, method) after 1, 5, 10 and 20 commits for contributors.

Component	**Number of First Commits**	**Categories**	**Founder No.**	**Joiner No.**	**Founder μ**	**Joiner μ**	**Welch Statistic**	**Welch P**	**MWU Statistic (Number of Pairs)**	**MWU P**
Packages	1	All	137	8303	0.44	0.10	2.76	0.0067	642,742.0 (1,137,511)	0.0000
Packages	1	Moderate	40	580	0.80	0.51	0.77	0.4477	12,483.0 (23,200)	0.2044
Packages	1	Sustained	34	203	0.38	0.65	-1.06	0.2921	3,461.5 (6,902)	0.9673
Packages	5	Moderate & sustained	74	783	3.36	3.85	-0.99	0.3232	29,616.5 (57,942)	0.7459
Packages	5	Moderate	40	580	3.52	3.72	-0.27	0.7874	12,075.0 (23,200)	0.6567
Packages	5	Sustained	34	203	3.18	4.22	-1.47	0.1459	3,369.0 (6,902)	0.8231
Packages	10	Moderate & sustained	74	783	5.07	6.05	-1.33	0.1855	28,593.0 (57,942)	0.8498
Packages	10	Moderate	40	580	5.15	6.09	-0.95	0.3462	11,610.5 (23,200)	0.9925
Packages	10	Sustained	34	203	4.97	5.94	-0.83	0.4084	3,298.5 (6,902)	0.6772
Packages	20	Sustained	34	203	7.18	9.62	-1.41	0.1628	3,181.5 (6,902)	0.4616
Classes	1	All	137	8303	4.01	0.64	2.92	0.0042	744,396.0 (1,137,511)	0.0000
Classes	1	Moderate	40	580	7.47	2.37	1.43	0.1603	12,255.5 (23,200)	0.4989
Classes	1	Sustained	34	203	2.85	4.81	-0.76	0.4494	3,783.0 (6,902)	0.2850
Classes	5	Moderate & sustained	74	783	39.05	24.11	1.77	0.0799	32,451.0 (57,942)	0.0872
Classes	5	Moderate	40	580	41.38	22.04	1.70	0.0969	13,640.0 (23,200)	0.0626
Classes	5	Sustained	34	203	36.32	30.01	0.48	0.6350	3,651.0 (6,902)	0.5895
Classes	10	Moderate & sustained	74	783	64.81	43.51	2.07	0.0419	32,846.5 (57,942)	0.0569
Classes	10	Moderate	40	580	61.60	41.95	1.40	0.1679	13,313.0 (23,200)	0.1180
Classes	10	Sustained	34	203	68.59	47.98	1.30	0.2003	3,879.0 (6,902)	0.2478
Classes	20	Sustained	34	203	107.50	81.91	1.34	0.1863	3,846.5 (6,902)	0.2857
Methods	1	All	137	8303	12.52	2.06	2.48	0.0144	743,810.0 (1,137,511)	0.0000
Methods	1	Moderate	40	580	24.27	7.22	1.32	0.1957	12,272.0 (23,200)	0.4881
Methods	1	Sustained	34	203	10.00	23.40	-0.77	0.4417	3,767.0 (6,902)	0.3092
Methods	5	Moderate & sustained	74	783	164.59	86.68	1.56	0.1218	32,324.5 (57,942)	0.0995
Methods	5	Moderate	40	580	186.75	76.85	1.46	0.1516	13,748.5 (23,200)	0.0499
Methods	5	Sustained	34	203	138.53	114.77	0.36	0.7211	3,550.5 (6,902)	0.7890
Methods	10	Moderate & sustained	74	783	263.26	154.41	1.76	0.0826	33,245.0 (57,942)	0.0358
Methods	10	Moderate	40	580	271.43	145.21	1.28	0.2063	13,417.5 (23,200)	0.0973
Methods	10	Sustained	34	203	253.65	180.70	1.00	0.3240	3,896.5 (6,902)	0.2291
Methods	20	Sustained	34	203	401.24	295.10	1.16	0.2533	3,693.5 (6,902)	0.5131

Table 3: Statistical significance tests of sample a category versus sample b category touches of different granularities of component (package, class, method) after 1, 5 and 10 commits for contributors.

Component	**Number of First Commits**	**Categories**	**Sample A No.**	**Sample B No.**	**Sample A** μ	**Sample B** μ	**Welch Statistic**	**Welch** P	**MWU Statistic (Number of Pairs)**	**MWU** P
Packages	1	transient versus moderate	7583	620	0.05	0.53	-4.79	0.0000	2,022,205.5 (4,701,460)	0.0000
Packages	1	transient versus sustained	7583	237	0.05	0.61	-3.62	0.0004	759,456.0 (1,797,171)	0.0000
Packages	1	moderate versus sustained	620	237	0.53	0.61	-0.43	0.6686	72,287.0 (146,940)	0.5695
Packages	5	moderate versus sustained	620	237	3.71	4.07	-0.80	0.4217	70,136.5 (146,940)	0.2931
Packages	10	moderate versus sustained	620	237	6.03	5.80	0.40	0.6905	73,098.0 (146,940)	0.9068
Classes	1	transient versus moderate	7583	620	0.41	2.70	-6.31	0.0000	1,608,530.5 (4,701,460)	0.0000
Classes	1	transient versus sustained	7583	237	0.41	4.53	-2.10	0.0368	674,951.0 (1,797,171)	0.0000
Classes	1	moderate versus sustained	620	237	2.70	4.53	-0.91	0.3614	78,310.0 (146,940)	0.0871
Classes	5	moderate versus sustained	620	237	23.29	30.92	-1.69	0.0919	74,229.5 (146,940)	0.8148
Classes	10	moderate versus sustained	620	237	43.22	50.94	-1.51	0.1316	74,941.0 (146,940)	0.6500
Methods	1	transient versus moderate	7583	620	1.12	8.32	-5.16	0.0000	1,609,157.0 (4,701,460)	0.0000
Methods	1	transient versus sustained	7583	237	1.12	21.48	-1.44	0.1522	674,687.5 (1,797,171)	0.0000
Methods	1	moderate versus sustained	620	237	8.32	21.48	-0.92	0.3564	78,436.0 (146,940)	0.0791
Methods	5	moderate versus sustained	620	237	83.94	118.18	-1.48	0.1389	73,502.5 (146,940)	0.9921
Methods	10	moderate versus sustained	620	237	153.35	191.16	-1.49	0.1366	73,214.0 (146,940)	0.9372

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{1}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1-1} + \frac{(s_2^2/N_2)^2}{N_2-1}} \tag{2}$$

1.7 Anova generate - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

1.7.1 Anova generate - For all components touched for a all commits excluding tranient

Table 4: Anova results for all components, number of commits and 8440 developers excluding transient

Component	Number of First Commits	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	1	Role (F/LJ)	3.14	1	3.31	0.0688	0.0004
packages	1	Category (T/M/S)	7.86	2	4.15	0.0158	0.0010
packages	1	Interaction	5.65	2	2.98	0.0509	0.0007
packages	1	Residual	7995.45	8434	-	-	
packages	5	Role (F/LJ)	1.43	1	0.05	0.8250	0.0001
packages	5	Category (M/S)	2.23	1	0.08	0.7826	0.0001
packages	5	Interaction	11.68	1	0.40	0.5279	0.0005
packages	5	Residual	24984.13	853	-	-	
packages	10	Role (F/LJ)	33.04	1	0.59	0.4443	0.0007
packages	10	Category (M/S)	0.59	1	0.01	0.9185	0.0000
packages	10	Interaction	0.01	1	0.00	0.9889	0.0000
packages	10	Residual	48127.58	853	-	-	
classes	1	Role (F/LJ)	973.61	1	24.93	0.0000	0.0029
classes	1	Category (T/M/S)	683.32	2	8.75	0.0002	0.0020
classes	1	Interaction	815.33	2	10.44	0.0000	0.0024
classes	1	Residual	329384.23	8434	-	-	
classes	5	Role (F/LJ)	13981.95	1	7.35	0.0069	0.0082
classes	5	Category (M/S)	468.97	1	0.25	0.6198	0.0003
classes	5	Interaction	2774.71	1	1.46	0.2276	0.0016
classes	5	Residual	1623543.63	853	-	-	
classes	10	Role (F/LJ)	14448.45	1	5.02	0.0254	0.0055
classes	10	Category (M/S)	897.52	1	0.31	0.5769	0.0003
classes	10	Interaction	15.03	1	0.01	0.9424	0.0000
classes	10	Residual	2457107.31	853	-	-	
methods	1	Role (F/LJ)	10883.40	1	6.87	0.0088	0.0008
methods	1	Category (T/M/S)	8092.96	2	2.56	0.0777	0.0006
methods	1	Interaction	15271.02	2	4.82	0.0081	0.0011
methods	1	Residual	13353418.84	8434	-	-	
methods	5	Role (F/LJ)	451979.70	1	8.27	0.0041	0.0093
methods	5	Category (M/S)	42733.87	1	0.78	0.3769	0.0009
methods	5	Interaction	121524.53	1	2.22	0.1364	0.0025
methods	5	Residual	46646591.43	853	-	-	
methods	10	Role (F/LJ)	596144.39	1	7.60	0.0060	0.0085
methods	10	Category (M/S)	5808.58	1	0.07	0.7856	0.0001
methods	10	Interaction	46476.28	1	0.59	0.4417	0.0007
methods	10	Residual	66927848.79	853	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{3}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{4}$$

1.8 Tukey hsd - For all components touched for all commits

An Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 5: Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	**Commits**	**Group A**	**Group B**	**Mean Diff**	**Adj *P* Value**
classes	1	Founder Moderate	Founder Sustained	-4.62	0.0190
classes	1	Founder Moderate	Founder Transient	-5.05	0.0009
classes	1	Founder Moderate	Late Joiner Moderate	-5.10	0.0000
classes	1	Founder Moderate	Late Joiner Transient	-7.08	0.0000
classes	1	Late Joiner Moderate	Late Joiner Sustained	2.43	0.0000
classes	1	Late Joiner Moderate	Late Joiner Transient	-1.98	0.0000
classes	1	Late Joiner Sustained	Late Joiner Transient	-4.42	0.0000
classes	5	Founder Moderate	Late Joiner Moderate	-19.33	0.0345
classes	10	Founder Sustained	Late Joiner Moderate	-26.64	0.0258
methods	1	Founder Moderate	Late Joiner Transient	-23.19	0.0032
methods	1	Founder Transient	Late Joiner Sustained	16.99	0.0364
methods	1	Late Joiner Moderate	Late Joiner Sustained	16.18	0.0000
methods	1	Late Joiner Moderate	Late Joiner Transient	-6.14	0.0046
methods	1	Late Joiner Sustained	Late Joiner Transient	-22.32	0.0000
methods	5	Founder Moderate	Late Joiner Moderate	-109.90	0.0215
methods	10	Founder Moderate	Late Joiner Moderate	-126.22	0.0303

Table 6: Not Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	**Commits**	**Group A**	**Group B**	**Mean Diff**	**Adj *P* Value**
classes	1	Founder Moderate	Late Joiner Sustained	-2.67	0.1341
classes	1	Founder Sustained	Founder Transient	-0.42	0.9996
classes	1	Founder Sustained	Late Joiner Moderate	-0.48	0.9981
classes	1	Founder Sustained	Late Joiner Sustained	1.95	0.5397
classes	1	Founder Sustained	Late Joiner Transient	-2.46	0.1975
classes	1	Founder Transient	Late Joiner Moderate	-0.05	1.0000
classes	1	Founder Transient	Late Joiner Sustained	2.38	0.0880
classes	1	Founder Transient	Late Joiner Transient	-2.04	0.1031
classes	5	Founder Moderate	Founder Sustained	-5.05	0.9599
classes	5	Founder Moderate	Late Joiner Sustained	-11.37	0.4344
classes	5	Founder Sustained	Late Joiner Moderate	-14.28	0.2486
classes	5	Founder Sustained	Late Joiner Sustained	-6.31	0.8631
classes	5	Late Joiner Moderate	Late Joiner Sustained	7.96	0.1137
classes	10	Founder Moderate	Founder Sustained	6.99	0.9443
classes	10	Founder Moderate	Late Joiner Moderate	-19.65	0.1135
classes	10	Founder Moderate	Late Joiner Sustained	-13.62	0.4581
classes	10	Founder Sustained	Late Joiner Sustained	-20.61	0.1631
classes	10	Late Joiner Moderate	Late Joiner Sustained	6.03	0.5137
classes	20	Founder Sustained	Late Joiner Sustained	-25.59	0.1034
methods	1	Founder Moderate	Founder Sustained	-14.28	0.6396
methods	1	Founder Moderate	Founder Transient	-17.86	0.2283
methods	1	Founder Moderate	Late Joiner Moderate	-17.05	0.0921
methods	1	Founder Moderate	Late Joiner Sustained	-0.88	1.0000
methods	1	Founder Sustained	Founder Transient	-3.59	0.9983
methods	1	Founder Sustained	Late Joiner Moderate	-2.78	0.9988
methods	1	Founder Sustained	Late Joiner Sustained	13.40	0.4546
methods	1	Founder Sustained	Late Joiner Transient	-8.92	0.7829
methods	1	Founder Transient	Late Joiner Moderate	0.81	1.0000
methods	1	Founder Transient	Late Joiner Transient	-5.33	0.8974
methods	5	Founder Moderate	Founder Sustained	-48.22	0.8132
methods	5	Founder Moderate	Late Joiner Sustained	-71.98	0.2840
methods	5	Founder Sustained	Late Joiner Moderate	-61.68	0.4410
methods	5	Founder Sustained	Late Joiner Sustained	-23.76	0.9470
methods	5	Late Joiner Moderate	Late Joiner Sustained	37.92	0.1931
methods	10	Founder Moderate	Founder Sustained	-17.78	0.9930
methods	10	Founder Moderate	Late Joiner Sustained	-90.73	0.2408
methods	10	Founder Sustained	Late Joiner Moderate	-108.44	0.1257
methods	10	Founder Sustained	Late Joiner Sustained	-72.95	0.4963
methods	10	Late Joiner Moderate	Late Joiner Sustained	35.49	0.4058
methods	20	Founder Sustained	Late Joiner Sustained	-106.14	0.1554

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{5}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df} \sqrt{\frac{MS_{residual}}{n}} \tag{6}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

1.9 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **50% Threshold:** The threshold number of components to touch to have a 50
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentate of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentate of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 7: Strategy Logistic Regression for all components against number of commits for 8440 developers excluding trainsient

Component	**Number of First Commits**	**Type**	**Moderate** μ	**Sustained** μ	**Coefficient**	**Odds Ratio**	**Std. Err**	p**-Value**	**30% Threshold**	**Moderate Met Threshold**	**Sustained Met Threshold**
packages	1	Intercept	0.40	3.99	-3.8260	0.0218	0.0752	0.0000	-	-	-
packages	1	Packages Count	0.40	3.99	0.2270	1.2548	0.0166	0.0000	**13.1243**	29 (0.4%)	17 (7.2%)
packages	5	Intercept	7.36	16.53	-1.3025	0.2719	0.0978	0.0000	-	-	-
packages	5	Packages Count	7.36	16.53	0.0324	1.0329	0.0055	0.0000	**14.0420**	116 (18.7%)	80 (33.8%)
packages	10	Intercept	9.68	23.99	-1.4121	0.2436	0.1019	0.0000	-	-	-
packages	10	Packages Count	9.68	23.99	0.0309	1.0313	0.0044	0.0000	**18.3064**	123 (19.8%)	90 (38.0%)
classes	1	Classes Count	2.82	36.49	0.0419	1.0428	0.0026	0.0000	**74.0370**	44 (0.5%)	28 (11.8%)
classes	5	Intercept	56.38	240.08	-1.8719	0.1538	0.1205	0.0000	-	-	-
classes	5	Classes Count	56.38	240.08	0.0092	1.0092	0.0009	0.0000	**111.4174**	86 (13.9%)	125 (52.7%)
classes	10	Intercept	79.55	407.66	-2.5977	0.0744	0.1560	0.0000	-	-	-
classes	10	Classes Count	79.55	407.66	0.0107	1.0108	0.0009	0.0000	**162.8939**	66 (10.6%)	160 (67.5%)
methods	1	Methods Count	9.44	133.10	0.0084	1.0084	0.0006	0.0000	**355.1713**	34 (0.4%)	16 (6.8%)
methods	5	Intercept	207.63	926.61	-1.6702	0.1882	0.1087	0.0000	-	-	-
methods	5	Methods Count	207.63	926.61	0.0019	1.0019	0.0002	0.0000	**438.0841**	67 (10.8%)	113 (47.7%)
methods	10	Intercept	289.62	1574.80	-2.2477	0.1056	0.1361	0.0000	-	-	-
methods	10	Methods Count	289.62	1574.80	0.0022	1.0022	0.0002	0.0000	**624.0901**	63 (10.2%)	150 (63.3%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon \tag{7}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{8}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.

2 Repository: 14

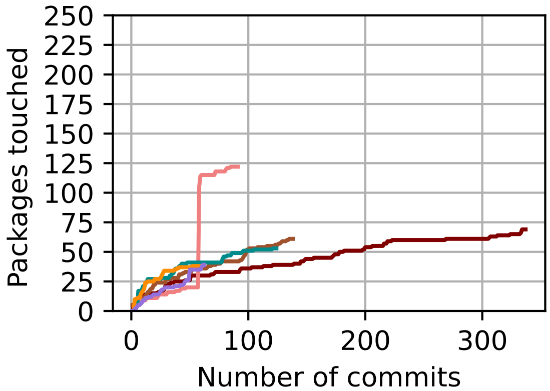
2.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 8: Table of first 6 sustained founder developers in the Scatter Developer graphs.

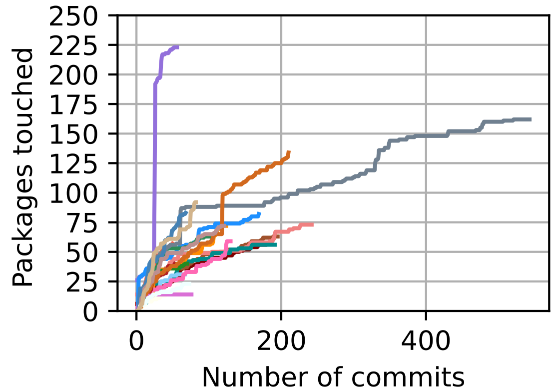
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 9: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

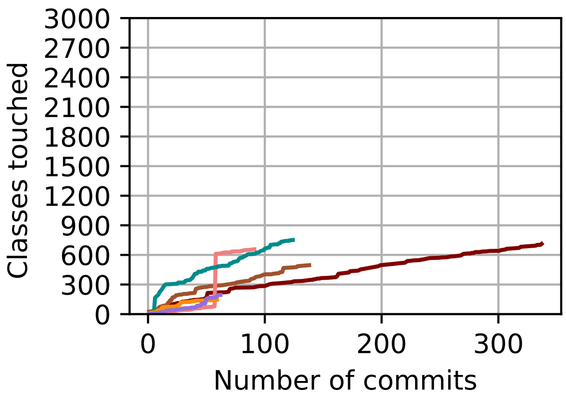
Developer ID	**Colour**
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



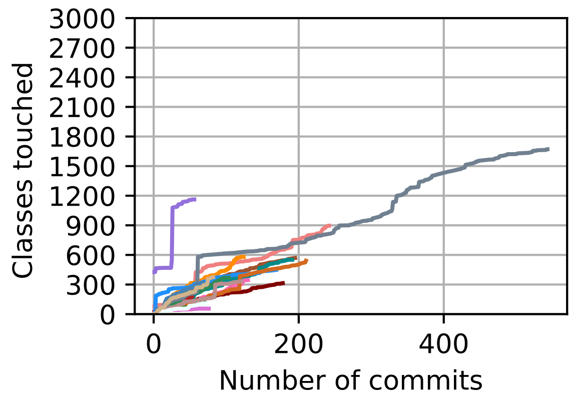
(a) Packages for sustained founder



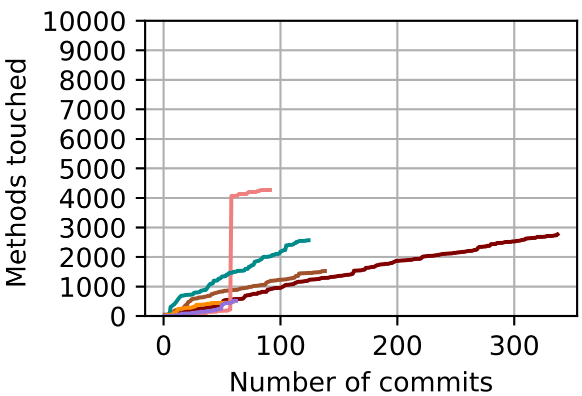
(b) Packages for sustained later joiner



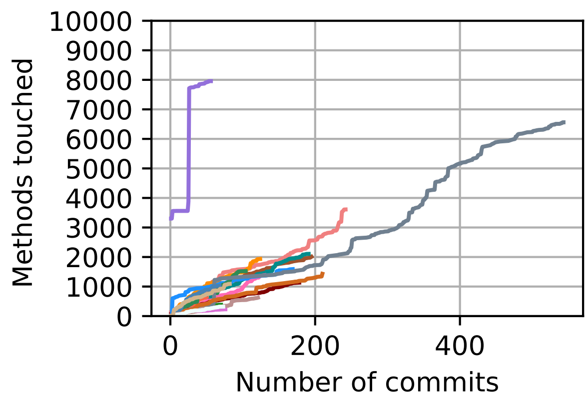
(c) Classes for sustained founder



(d) Classes for sustained later joiner



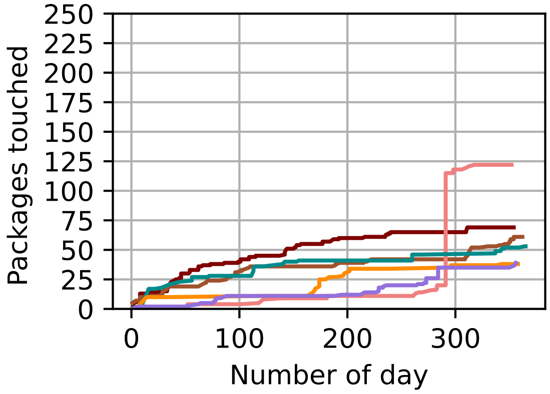
(e) Methods for sustained founder



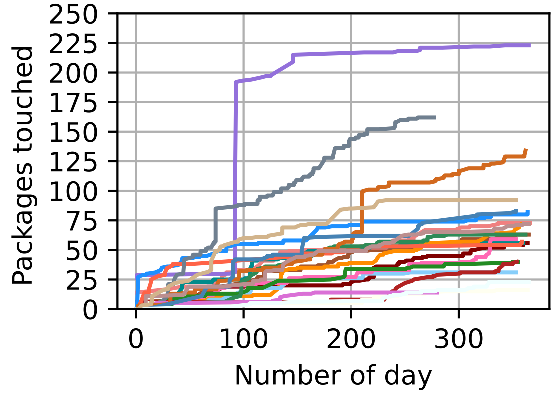
(f) Methods for sustained later joiner

Figure 32: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 8. Sustained later joiner developers with colours are shown in Table 9.

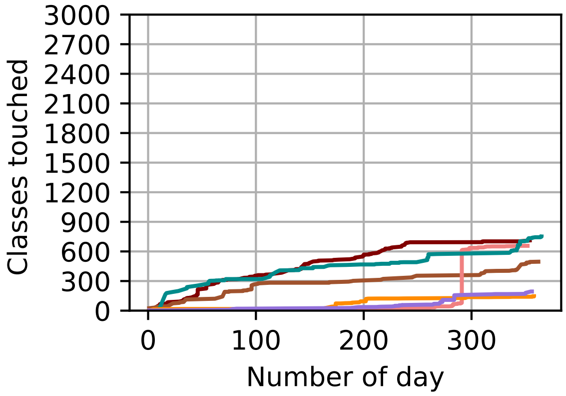
2.2 Time series sustained components touched by developer - 14 For developer day and components touched.



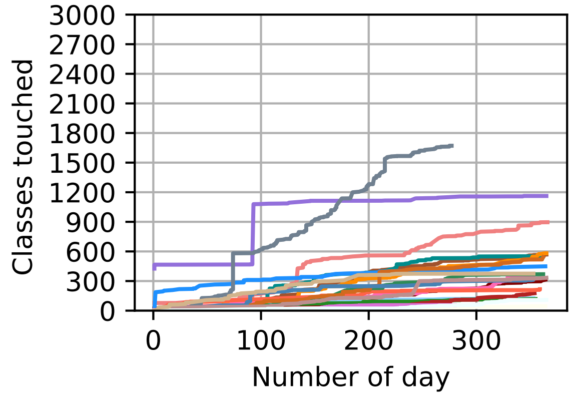
(a) Packages for sustained founder



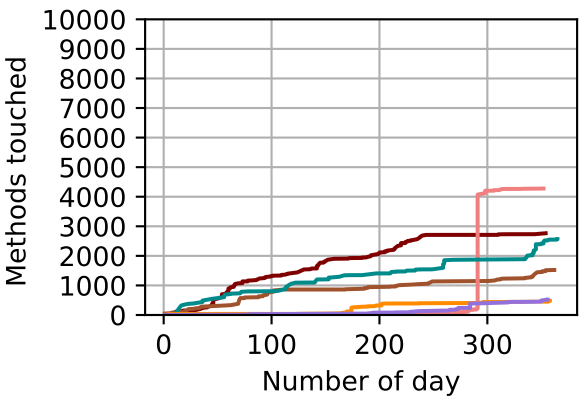
(b) Packages for sustained later joiner



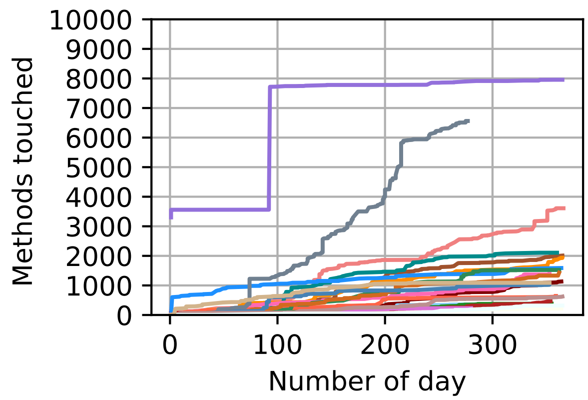
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 33: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 8. Sustained later joiner developers with colours are shown in Table 9.

3 Repository: 21

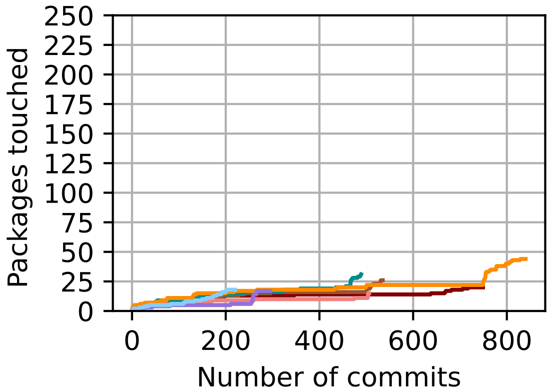
3.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 10: Table of first 7 sustained founder developers in the Scatter Developer graphs.

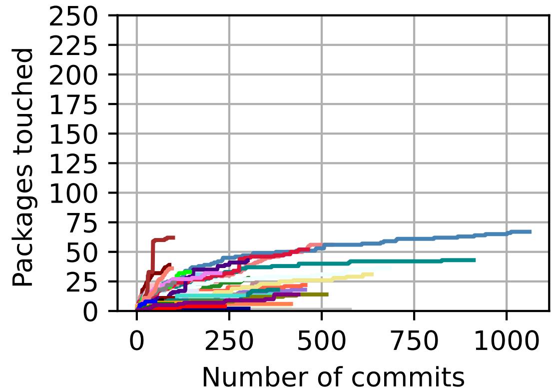
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 11: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

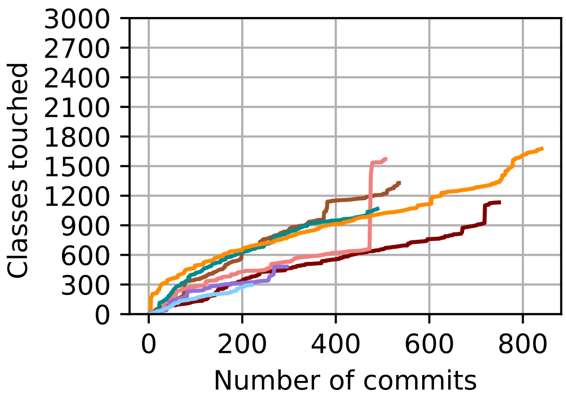
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



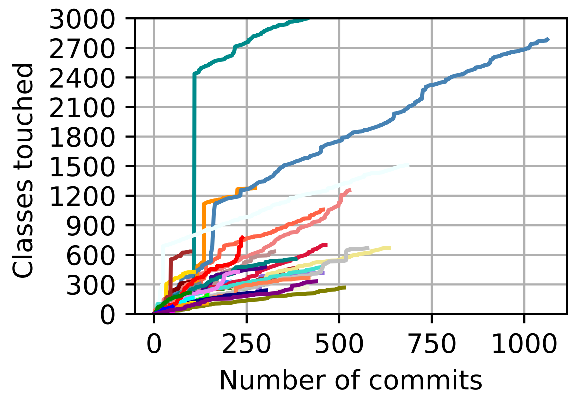
(a) Packages for sustained founder



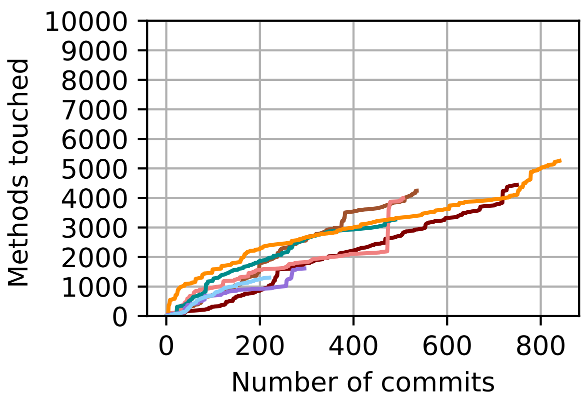
(b) Packages for sustained later joiner



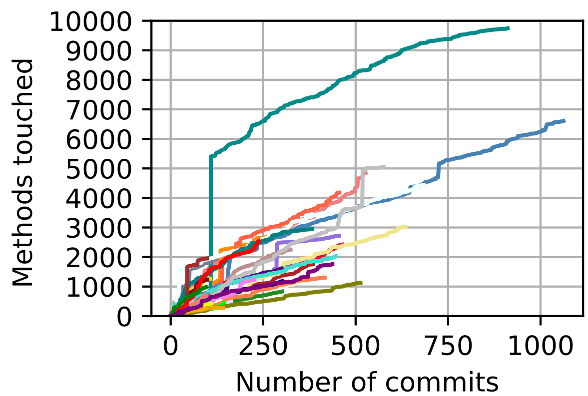
(c) Classes for sustained founder



(d) Classes for sustained later joiner



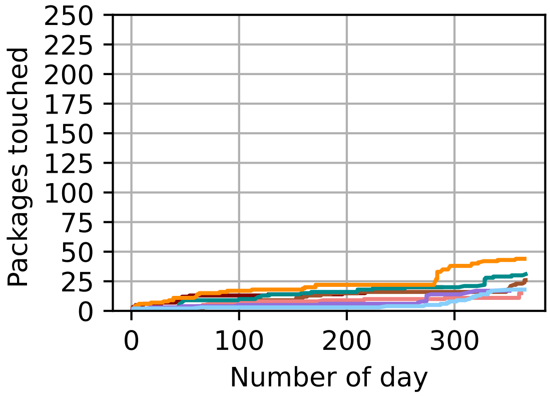
(e) Methods for sustained founder



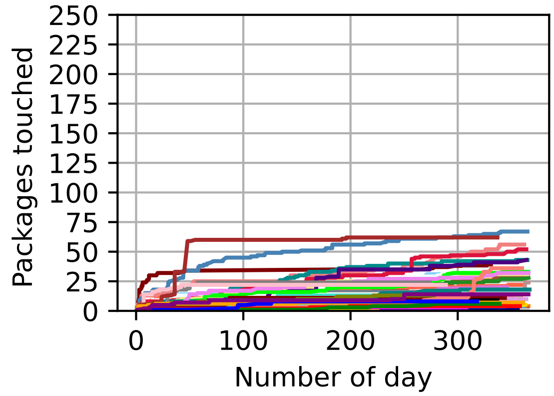
(f) Methods for sustained later joiner

Figure 34: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 10. Sustained later joiner developers with colours are shown in Table 11.

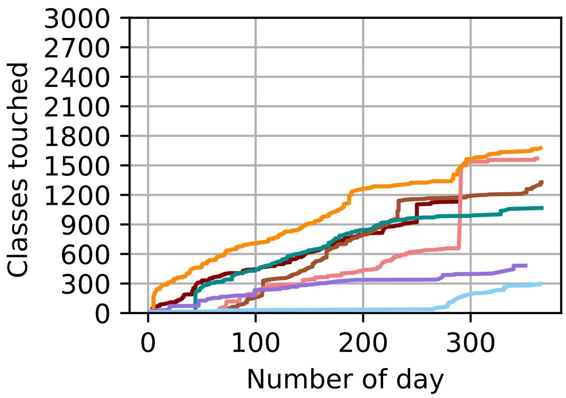
3.2 Time series sustained components touched by developer - 21 For developer day and components touched.



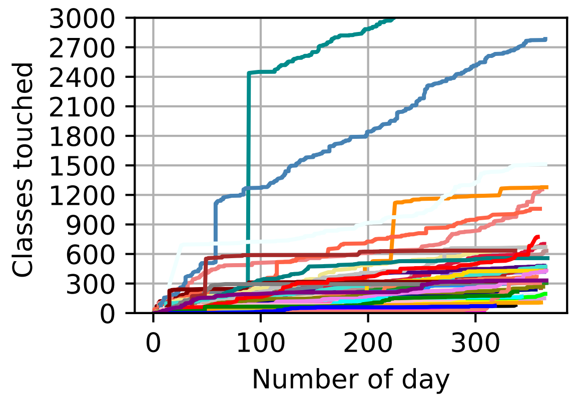
(a) Packages for sustained founder



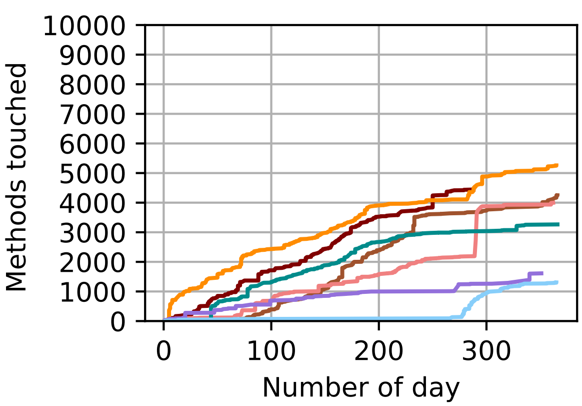
(b) Packages for sustained later joiner



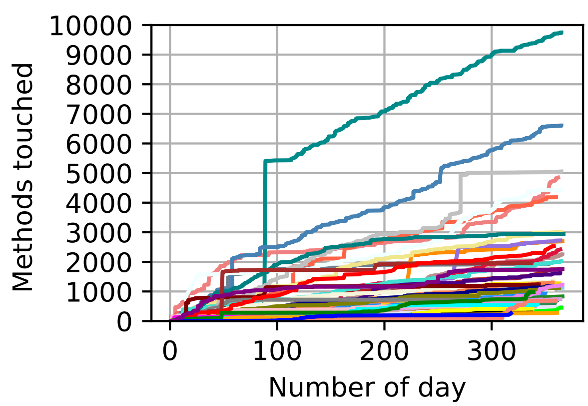
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 35: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 10. Sustained later joiner developers with colours are shown in Table 11.